Trust-Aware Multi-Agent Reinforcement Learning-Based Cooperative Interception Strategy for Maneuvering Swarms with Target Assignment

***Abstract*—The swarm attack of unmanned aerial vehicles (UAVs) poses a significant threat to future battlefield operations. Achieving efficient and precise cooperative interception against attacking UAV swarms with highly dynamic maneuvering capabilities and adaptive penetration strategies poses a critical challenge for UAV interceptor swarms. This study addresses the cooperative interception of maneuvering targets in swarm attacks by decomposing the problem into two key tasks: target assignment and cooperative interception guidance. For target assignment, an Improved Dung Beetle Optimization (IDBO) algorithm is proposed to dynamically balance exploration and exploitation, achieving highly efficient and rapid assignments. For cooperative interception guidance, an Attention-Residual and Trust-aware Multi-Agent Proximal Policy Optimization (ART-MAPPO) framework is developed to achieve precise and time-coordinated interception of adversarial UAV swarms employing advanced penetration strategies. The simulation results validate the effectiveness of the proposed intelligent cooperative interception method, demonstrating superior convergence speed, robustness, and significantly improved interception success rates in swarm UAV cooperative interception scenarios.**

***Note to Practitioners*—This paper is motivated by practical UAV swarm defense scenarios, where multiple interceptors must cooperatively engage maneuvering aerial targets under limited communication and onboard maneuver constraints. In such applications, target assignment and guidance are closely coupled, and conventional centralized approaches may be difficult to implement efficiently as the engagement scale increases. To address this issue, this paper develops a two-stage cooperative interception framework that combines distributed target assignment with a trust-aware multi-agent reinforcement learning guidance method. The proposed approach is intended to improve coordination consistency among interceptors assigned to the same target and to enhance robustness against target maneuvers while considering maneuver saturation and command smoothness. This work may be useful for practical large-scale multi-UAV interception and other cooperative autonomous systems requiring scalable distributed decision-making and coordinated guidance.**

***Index Terms*—Cooperative interception, deep reinforcement learning, centralized training with decentralized execution, target assignment, UAV swarm.**

I. Introduction

As warfare evolves toward intelligent swarm operations, the likelihood of encountering attacks from adversarial UAV swarms significantly increases during combat. Consequently, the efficient interception of incoming swarm targets becomes a critical imperative for future battlefield operations. To address the challenge of cooperative interception against maneuvering attacking swarms with adaptive penetration strategies, there is an urgent need to advance academic research in swarm target interception technologies. This study aims to enhance interceptors with intelligent cooperative interception capabilities, enabling effective navigation of complex challenges in modern battlefield environments.

Multi-target cooperative interception encompasses two key tasks: target assignment and cooperative interception. The effectiveness of multi-target interception hinges on the optimality of target assignment, establishing it as a primary focus in swarm target interception strategies. As the number of UAVs increases, the complexity of target assignment escalates significantly, resulting in a multi-constrained, multi-parameter optimization problem. The objective is to develop a rapid and efficient target assignment method for allocating specific attacking targets to interceptors in operational scenarios [?]. The target assignment problem involves complex constraints, and the non-dominated sorting algorithm can dynamically generate a set of reference points with good convergence and distribution performance [?], [?], [?]. Moreover, employing an online operator selection mechanism can enhance the efficiency of the algorithm [?], [?]. The target assignment problem can be solved using multi-information particle swarm optimization to find the solution that yields the highest gain [?], [?]. Introducing information from surrounding particles can enhance the stability of the traditional particle swarm algorithm while also addressing the issue of local convergence in the target assignment problem [?]. The genetic algorithm is a common approach for solving the target assignment problem [?], [?], [?], [?], and a GA with a novel crossover operator can further enhance algorithmic efficiency [?]. Ant colony optimization is also an effective strategy for solving the target assignment problem [?], and its combination with a greedy algorithm can improve the accuracy of parallel computation [?]. Under multiple constraints, researchers have proposed various heuristic algorithms to solve the target assignment problem, such as neural networks [?], auction algorithm [?], tabu search [?], simulated annealing algorithm [?], VLSN [?]. However, in cooperative interception scenarios, the aforementioned algorithms struggle to handle the highly dynamic nature of the battlefield due to their high time complexity, and they fail to meet the dual requirements of both speed and optimality.

The effectiveness of the terminal guidance phase in multi-target interception is significantly influenced by the cooperative guidance law, a critical factor requiring thorough consideration. The target assignment strategy transforms swarm operations into a coordinated multi-to-one interception problem, with current research focusing on three key areas: angular coordination, temporal coordination, and intelligent cooperation. Significant progress has been made in temporal coordination. Classical proportional navigation-based guidance laws enable synchronized multi-missile attacks on a single target [?]. Adaptive control methods adjust attack timing based on real-time operational conditions, making them suitable for intercepting maneuvering targets [?], [?]. Sliding Mode Control (SMC) is widely adopted for cooperative interception [?], with SMC incorporating miss distance and attack time constraints enhancing interception accuracy [?], [?], [?]. Considerable research has also been conducted on angular coordination. From a traditional control perspective, backstepping control [?] and leader-follower cooperative methods [?] significantly improve interception efficiency. SMC with angular constraints is prevalent in cooperative interception, encompassing approaches based on impact angle constraints [?] and line-of-sight angle constraints [?]. Optimal control methods applied to impact angle-constrained cooperative guidance enhance precision by integrating impact angle and line-of-sight angle constraints [?], [?].

Recent advancements in intelligent technologies have driven extensive research into cooperative interception strategies for UAV swarms, with intelligent algorithms demonstrating substantial potential in addressing complex multi-target scenarios [?], [?]. Optimization-based approaches, such as Particle Swarm Optimization (PSO), excel in collaborative attack problems by autonomously optimizing resource allocation and maximizing operational efficiency [?], [?]. Search-based algorithms, like A* [?], enable efficient path planning for coordinated attacks. Control-based methods, including Model Predictive Control (MPC) [?], provide robust guidance under dynamic constraints. Moreover, the integration of neural networks with reinforcement learning, particularly deep reinforcement learning (DRL), enhances the adaptability and intelligence of interception strategies [?], [?], [?]. DRL enables UAV swarms to learn complex policies from high-dimensional data, optimizing target assignment and maneuvering in real-time to counter maneuvering targets [?]. In contrast, conventional cooperative interception guidance methods rely heavily on linearization and simplifying assumptions to design feasible guidance laws [?]. These methods often struggle to address the complexity of dynamic, non-linear environments involving maneuvering swarm targets. Data-driven intelligent cooperative interception methods, by contrast, leverage extensive datasets to learn adaptive strategies, significantly reducing dependence on restrictive assumptions [?]. By modeling complex interactions through neural networks and DRL, these methods achieve greater robustness and flexibility. Despite advancements, intelligent cooperative interception methods struggle with maneuvering swarm targets in complex dynamic environments due to limited model generalization, low training efficiency, and low interception success rates, particularly in large-scale swarm combat requiring precise temporal coordination [?]. Addressing these challenges requires further research into scalable, generalizable, and efficient algorithms to fully realize the potential of intelligent cooperative interception.

In summary, existing cooperative interception methods for countering attacking UAV swarms with adaptive penetration capabilities suffer from low interception success rates in complex dynamic environments. Specifically, target assignment methods exhibit low allocation efficiency, while cooperative interception guidance strategies struggle with terminal overload saturation and insufficient adaptability to dynamic target maneuvers in practical scenarios, hindering effective swarm coordination. Consequently, more research is necessary to address these challenges.

This study proposes a swarm cooperative interception method based on multi-agent reinforcement learning and IDBO algorithm, enhancing target assignment efficiency and guidance strategy performance to achieve precise and efficient interception of maneuvering targets. The proposed intelligent cooperative interception algorithm offers the following advantages.

1) A distributed consensus-based target assignment scheme is proposed, which integrates an improved Dung Beetle Optimization (IDBO) algorithm with a combat-dominance assessment model. This design supports rapid many-to-one allocation under limited communication, leading to more efficient and reliable target assignment for cooperative interception.
2) ART-MAPPO is proposed as a CTDE-based cooperative interception guidance learning framework, which employs an attention–residual feature extractor and a GRU temporal encoder, and uses trust-modulated exploration together with dual-clip PPO and adaptive KL regularization for stable policy learning. This design improves convergence behavior and training efficiency in highly maneuverable swarm-adversarial environments.
3) A task-oriented reward design and training stabilization strategy are incorporated into the cooperative interception framework to support time-coordinated guidance under bounded overload commands. Rather than treating reward shaping or dual-clip PPO as standalone methodological novelties, this work uses them as task-adapted components that couple target approach, terminal interception, temporal coordination, and control-effort regularization in a large-scale maneuvering swarm scenario.

The paper is structured as follows: Section **??** presents relevant problem modeling and foundational theory. Section **??** and Section **??** provide a detailed description of our approach, including its framework, modules, algorithms, and rationale. Section **??** covers experiments, results, and discussions. Lastly, Section **??** summarizes the paper and outlines future directions.

II. Backgrounds and Preliminaries

In this section, we first introduce the modeling of multi-target interception of UAVs, and then review reinforcement learning methods.

A. Problem Modeling

The multi-target intelligent interception problem can be decomposed into two subproblems: target assignment and intelligent cooperative interception, with the swarm combat

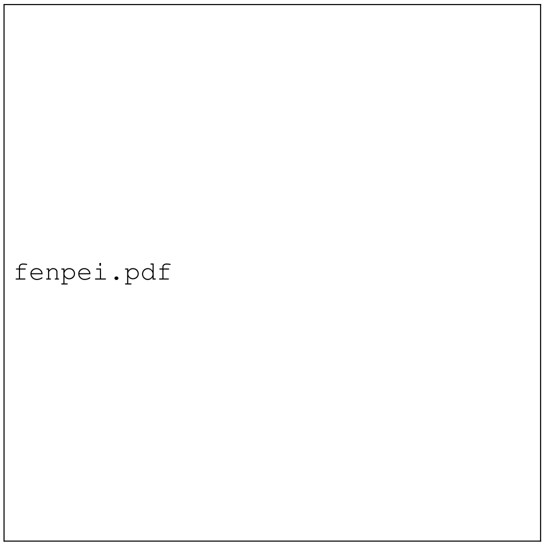


Fig. 1. Cluster target interception problem.

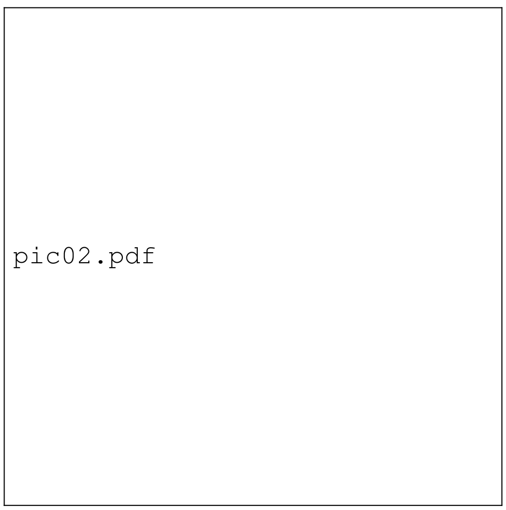


Fig. 2. Schematic diagram of confrontation and interception.

scenario illustrated in Fig. **??**. In this scenario, the attacking swarm aims to penetrate a high-value target area, while the defending swarm seeks to fully intercept the incoming UAVs. During the target assignment phase, the defending swarm assesses its own flight status and that of the attacking swarm using onboard sensors, subsequently allocating each defender to an appropriate interception target via the target assignment strategy. In the intelligent cooperative interception phase, defenders assigned to the same target employ coordinated maneuvering strategies to intercept and neutralize the target.

The object of study in this paper is the fixed-wing UAV. Fig. **??** shows the horizontal projection used to define the heading and line-of-sight (LOS) azimuth, whereas the guidance model and simulations propagate the complete three-dimensional position and velocity.

The three-dimensional point-mass kinematic model of a single UAV is

$$\begin{cases} \dot{x} = V\cos\theta\cos\gamma, \\ \dot{y} = V\cos\theta\sin\gamma, \\ \dot{z} = V\sin\theta, \end{cases} \tag{1}$$

where (x, y, z) are the three-dimensional position coordinates, V is the velocity magnitude, γ is the heading angle, and θ is the flight-path (pitch) angle.

In an interception mission, let $\mathbf{p}_A, \mathbf{p}_D \in \mathbb{R}^3$ and $\mathbf{v}_A, \mathbf{v}_D \in \mathbb{R}^3$ denote the attacker and defender states. Their three-dimensional relative geometry is

$$\begin{aligned} \mathbf{r}_{AD} &= \mathbf{p}_A - \mathbf{p}_D, & r &= \|\mathbf{r}_{AD}\|_2, \\ \boldsymbol{\ell}_{AD} &= \mathbf{r}_{AD}/r, & \dot{r} &= \boldsymbol{\ell}_{AD}^{\top}(\mathbf{v}_A - \mathbf{v}_D). \end{aligned} \tag{2}$$

Here, r is the three-dimensional inter-UAV distance and $\boldsymbol{\ell}_{AD}$ is the LOS unit vector; its horizontal projection provides the plan-view geometry shown in Fig. **??**.

The relationship between the flight state and the three commanded load channels is

$$\begin{cases} \dot{V} = n_x g, \\ \dot{\gamma} = \dfrac{n_y g}{V\cos\theta}, \\ \dot{\theta} = \dfrac{n_z g}{V}, \end{cases} \tag{3}$$

where n_x is the axial load-factor command, n_y and n_z are the horizontal and vertical normal load-factor commands, respectively, and g is gravitational acceleration. Thus, n_y primarily changes heading, while n_z changes the flight-path angle and altitude. In the multi-agent setting of Section **??**, these variables carry an agent subscript i.

During cooperative interception, the remaining impact time is defined as:

$$t_{go} = \frac{r}{|\dot{r}|} \tag{4}$$

The three-dimensional Zero-Effort Miss (ZEM) is defined as

$$\mathrm{ZEM} = \left\|\mathbf{r}_{AD} + (\mathbf{v}_A - \mathbf{v}_D)t_{go}\right\|_2. \tag{5}$$

B. Reinforcement Learning

Reinforcement learning (RL) is built on the Markov Decision Process (MDP), comprising state space $\mathcal{S}$, action space $\mathcal{A}$, reward function R, policy π, and state transition function P. At each step, the agent selects action a from state s, receives reward r, and transitions to s'. The key value functions are:

$$V^{\pi}(s) = E_{\pi}\left[\sum_{t=0}^{\infty} \gamma_{\mathrm{RL}}^{t} r_t \mid s_0 = s\right] \tag{6}$$

$$Q^{\pi}(s,a) = E_{\pi}\left[\sum_{t=0}^{\infty} \gamma_{\mathrm{RL}}^{t} r_t \mid s_0 = s, a_0 = a\right] \tag{7}$$

where $\gamma_{\mathrm{RL}} \in [0,1]$ is the discount factor.

Fig. 3. Bipartite representation of defender–target assignment.

III. Target Assignment via Distributed Consensus-Based IDBO Algorithm

This section formulates cooperative target assignment as a capacitated bipartite matching problem and then describes how local IDBO search and neighbor-to-neighbor bidding produce a consistent many-to-one assignment. To avoid overloading the notation used later for learning, let $\mathcal{D} = \{d_1, \dots, d_{N_D}\}$ and $\mathcal{T} = \{t_1, \dots, t_{N_A}\}$ denote the defender and target sets, respectively; i always indexes a defender and j a target in this section.

A. Pairwise Interception Model and Assignment Objective

Figure **??** illustrates the bipartite assignment structure. The binary variable $X_{ij} = 1$ indicates that defender d_i is assigned to target t_j. A useful edge should simultaneously have a small predicted miss tendency and a favorable heading geometry. We therefore define the pairwise interception-probability proxy as

$$
\begin{aligned}
p_{ij}^{\text{int}} &= \omega_Z \exp\left[-\frac{1}{2}\left(\frac{\text{ZEM}_{ij}}{s_Z}\right)^2\right] + \omega_\ell \exp\left[-\frac{1}{2}\left(\frac{e_{ij}^{\ell}}{s_\ell}\right)^2\right], \\
e_{ij}^{\ell} &= \arccos\left(\text{clip}\left(\frac{\mathbf{v}_{D_i}^{\top}\boldsymbol{\ell}_{ij}}{\|\mathbf{v}_{D_i}\|_2 + \varepsilon_{\text{num}}}, -1, 1\right)\right),
\end{aligned}
\tag{8}
$$

where $s_Z = [(N_D N_A)^{-1}\sum_{i,j}\text{ZEM}_{ij}^2]^{1/2} + \varepsilon_{\text{num}}$ and $s_\ell = [(N_D N_A)^{-1}\sum_{i,j}(e_{ij}^{\ell})^2]^{1/2} + \varepsilon_{\text{num}}$ are root-mean-square normalization scales. Here, $\boldsymbol{\ell}_{ij} = (\mathbf{p}_{T_j} - \mathbf{p}_{D_i})/\|\mathbf{p}_{T_j} - \mathbf{p}_{D_i}\|_2$ is the three-dimensional line-of-sight unit vector, and $\text{clip}(\cdot, -1, 1)$ keeps the inverse-cosine argument numerically valid. The nonnegative weights satisfy $\omega_Z + \omega_\ell = 1$. Thus, either a large predicted miss or poor velocity-to-LOS alignment reduces the edge value, whereas a geometrically favorable pair has p_{ij}^{int} close to one. The RMS scales avoid cancellation of signed ZEM values and make the two terms comparable. In implementation, this bounded quantity is used as a probability proxy; the product model below adopts the usual conditional-independence approximation for different defenders assigned to the same target.

The tactical objective is to reduce the expected number of surviving targets. Assigning an additional capable defender to target t_j multiplies its approximate survival probability by another failure factor. The resulting capacitated assignment is

$$
\begin{aligned}
\min_{\mathbf{X}} \quad & J(\mathbf{X}) = \sum_{j=1}^{N_A}\prod_{i=1}^{N_D}\left(1 - p_{ij}^{\text{int}}\right)^{X_{ij}} \\
\text{s.t.} \quad & \sum_{j=1}^{N_A} X_{ij} = 1, && i = 1, \dots, N_D, \\
& 1 \le \sum_{i=1}^{N_D} X_{ij} \le L_{\max}, && j = 1, \dots, N_A, \\
& X_{ij} \in \{0, 1\}.
\end{aligned}
\tag{9}
$$

The first constraint assigns every defender to exactly one target; the second prevents both uncovered targets and excessive concentration. A feasible assignment requires $N_A \le N_D \le L_{\max} N_A$.

B. Local IDBO Search and Consensus Bidding

The distributed solution alternates between two interpretable operations. Each defender first searches for a promising target using locally available engagement information; neighboring defenders then reconcile conflicting choices by exchanging target-specific bids. Defender d_i maintains an unconstrained preference vector $\boldsymbol{\xi}_i \in \mathbb{R}^{N_A}$ and maps it to soft preferences $\pi_{ij} = \sigma(\xi_{ij})$, where $\sigma(x) = (1 + e^{-x})^{-1}$. This separates the continuous search variable ξ_{ij} from the binary assignment variable X_{ij}.

For target t_j, let $\widehat{n}_{ij} = \sum_{\ell \in \mathcal{N}_i \setminus \{i\}} \widehat{\pi}_{\ell j}$ be the assignment load estimated from defender d_i's neighbor set $\mathcal{N}_i$. The target-specific utility and aggregate local fitness are

$$
\begin{aligned}
u_{ij}(\boldsymbol{\xi}_i) &= \pi_{ij}\left(p_{ij}^{\text{int}} + \lambda_A \chi_{ij}\right) - \lambda_L \pi_{ij}\left[\widehat{n}_{ij} + \pi_{ij} - L_{\max}\right]_+, \\
\mathcal{F}_i(\boldsymbol{\xi}_i) &= \sum_{j=1}^{N_A} u_{ij}(\boldsymbol{\xi}_i),
\end{aligned}
\tag{10}
$$

where $[x]_+ = \max(0, x)$ and χ_{ij} is the combat advantage. The first term favors interceptable and tactically advantageous targets; the hinge term is inactive below the estimated capacity and increases only when another defender would oversubscribe that target.

The advantage combines time, relative-velocity compatibility, encounter geometry, and local competition:

$$
\begin{aligned}
\chi_{ij} = {} & \exp\left(-\frac{\Delta t_{ij}}{T_c}\right)\left[1 - \frac{\|\mathbf{v}_{D_i} - \mathbf{v}_{T_j}\|_2}{v_c}\right]_+ [\cos(\vartheta_{ij})]_+ \\
& - \lambda_C \sum_{\ell \in \mathcal{N}_i \setminus \{i\}} \frac{p_{\ell j}^{\text{int}}}{p_{ij}^{\text{int}} + p_{\ell j}^{\text{int}} + \varepsilon_{\text{num}}}\widehat{\pi}_{\ell j}.
\end{aligned}
\tag{11}
$$

Here, Δt_{ij} is the estimated intercept time, ϑ_{ij} is the encounter aspect angle, and T_c and v_c are positive normalization constants. The positive-part operators prevent a large relative-speed mismatch or an unfavorable aspect angle from producing a negative multiplicative "advantage." The final term lowers the utility of a target already preferred by capable neighbors.

Rather than presenting four long update equations in succession, we collect the rolling, dancing, breeding, and stealing operations into one implementation operator:

$$\boldsymbol{\xi}_i^{(k+1)} = \Pi_{\Xi}\Big[\mathcal{U}_{\mathrm{IDBO}}\big(\boldsymbol{\xi}_i^{(k)}, \mathcal{F}_i, \boldsymbol{\chi}_i^{(k)}; k\big)\Big]. \tag{12}$$

The four operations respectively perform local exploitation, controlled perturbation, elite recombination, and preference exchange. Their coefficients decay with assignment iteration k, so early iterations explore broadly and later iterations refine high-utility candidates. Π_{Ξ} clips the latent preferences to their numerical search bounds after every update. The procedural order and feasibility repair are stated explicitly in Algorithm **??**; the compact form in (**??**) keeps the main physical flow visible.

After local search, defender d_i selects exactly one candidate target,

$$\widetilde{X}_{ij}^{(k+1)} = \mathbb{I}\left[j = \underset{r\in\{1,\ldots,N_A\}}{\arg\max}\ u_{ir}(\boldsymbol{\xi}_i^{(k+1)})\right], \tag{13}$$

with deterministic index-based tie breaking. Unlike independent thresholding, this mapping cannot return either zero targets or multiple targets for one defender. The corresponding bid amplifies targets for which the defender has an above-average combat advantage:

$$b_{ij}^{(k+1)} = \widetilde{X}_{ij}^{(k+1)} u_{ij} \exp\left(\frac{\chi_{ij} - \bar{\chi}_i}{s_{\chi,i} + \varepsilon_{\mathrm{num}}}\right), \tag{14}$$

where $\bar{\chi}_i$ and $s_{\chi,i}$ are the mean and standard deviation of $\{\chi_{ij}\}_{j=1}^{N_A}$.

Each defender stores a local winner set $\mathcal{W}_{i,j}$ for every target. After receiving neighbor copies, it retains the highest $L_{\max}$ bids:

$$\mathcal{W}_{i,j}^{(k+1)} = \mathrm{Top}_{L_{\max}}\left(\mathcal{W}_{i,j}^{(k)} \cup \bigcup_{\ell\in\mathcal{N}_i} \mathcal{W}_{\ell,j}^{(k)} \cup \{(d_i, b_{ij}^{(k+1)})\}\right). \tag{15}$$

Because only the selected target receives a nonzero bid, the winner sets enforce the target capacity without destroying the one-target-per-defender interpretation. A defender displaced from a full winner set removes that target from its local candidate and searches/bids again in the next iteration.

Let $\mathbf{W}_i^{(k)} = \{\mathcal{W}_{i,j}^{(k)}\}_{j=1}^{N_A}$ and let $\mathcal{E}$ denote the communication edges. The normalized pairwise disagreement is

$$\Gamma^{(k)} = \frac{1}{|\mathcal{E}|} \sum_{(i,\ell)\in\mathcal{E}} \mathbb{I}\Big[\mathbf{W}_i^{(k)} \neq \mathbf{W}_\ell^{(k)}\Big] \leq \varepsilon_{\mathrm{con}}. \tag{16}$$

This definition directly counts communication links whose winner information is inconsistent; it is zero when all neighboring defenders agree.

C. Convergence Scope and Computational Cost

The convergence statement concerns agreement of the distributed assignment process for a fixed engagement snapshot, not global optimality of the nonconvex problem in (**??**). We assume that the defender/target sets are fixed during one assignment round, the communication graph is connected with bounded delay, bids use deterministic tie breaking, and

assignment.pdf

Fig. 4. Distributed IDBO assignment flow: local preference search, one-hot candidate selection, top-$L_{\max}$ consensus bidding, and engagement-information update.

accepted local candidates are retained only when they improve the projected local fitness. Under these conditions, each target's highest bids propagate through the connected graph in finitely many communication rounds. Since the binary feasible set and the winner sets are finite, the deterministic local-search/consensus process reaches a locally stable fixed point unless the prescribed iteration limit is met first. This guarantees consistent winner information; it does not establish a global optimum or an ε-Nash bound.

For one defender and one assignment iteration, evaluating a population of N_{pop} candidates over N_A targets costs $O(N_{\mathrm{pop}}N_A)$, exchanging target bids costs $O(N_A|\mathcal{N}_i|)$, and updating the advantage costs $O(N_A)$. For K_{IDBO} iterations, these terms are multiplied by K_{IDBO}; per-defender memory is $O(N_{\mathrm{pop}}N_A + N_A)$.

IV. ART-MAPPO: Attention-Residual and Trust-Aware MAPPO for Cooperative Swarm Interception

Given the consensus assignment $\mathbf{X}^*$, ART-MAPPO learns cooperative guidance under the centralized-training–decentralized-execution (CTDE) paradigm. Its information flow is kept explicit throughout this section: normalized physical observations are encoded by a task-aware attention-residual backbone, a GRU retains engagement history, trust modulates training-time exploration, and dual-clip/KL regularization stabilizes the policy update.

A. Task-Aware Attention-Residual Policy Network

Let $\mathbf{o}_i^{(t)} = [o_i^{(t),1}, \ldots, o_i^{(t),n_o}]^\top$ denote defender i's local observation at guidance step t. Instead of applying a fully

Algorithm 1 Distributed Consensus-Based IDBO Target Assignment

Require: $\mathcal{D}$, $\mathcal{T}$, p_{ij}^{int}, neighbor sets $\mathcal{N}_i$, capacity $L_{\max}$, tolerance ε_{con}, maximum iteration $K_{\max}$
Ensure: Consensus assignment $\mathbf{X}^*$

```
Initialize ξ_i, bids b_ij, and local winner sets W_{i,j} for every defender
for k = 0 to K_max − 1 do
    for each defender d_i do
        Evaluate p_ij^int, χ_ij, and u_ij from the current engagement snapshot
        Update ξ_i with the IDBO operator in (??) and project it to the search bounds
        Form the one-hot candidate X̃_{i,:} using (??) and compute its bid using (??)
        Exchange {W_{i,j}}_{j=1}^{N_A} with neighbors and retain top-L_max bids using (??)
        If displaced, remove the conflicting candidate before the next search step
    end for
    Compute Γ^(k+1) using (??)
    if Γ^(k+1) ≤ ε_con then
        break
    end if
end for
Construct X* from the agreed winner sets and apply a deterministic feasibility repair if an uncovered target remains
```

connected layer and then arbitrarily reshaping mixed features into tokens, we embed each physical channel separately:

$$\mathbf{H}_i^{(0)} = \left[f_1(o_i^{(t),1})^\top \quad \cdots \quad f_{n_o}(o_i^{(t),n_o})^\top \right]^\top \in \mathbb{R}^{n_o \times d_e}, \tag{17}$$

where each $f_r : \mathbb{R} \to \mathbb{R}^{d_e}$ is a learned channel embedding. Consequently, one token continues to represent one interpretable guidance quantity, such as LOS rate or time-coordination error.

For attention head $h = 1, \dots, H$, let $d_h = d_e/H$. The complete attention mapping is

$$\begin{aligned} \mathbf{Q}_h &= \mathbf{H}_i^{(0)}\mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{H}_i^{(0)}\mathbf{W}_h^K, \quad \mathbf{U}_h = \mathbf{H}_i^{(0)}\mathbf{W}_h^U, \\ \boldsymbol{\alpha}_h &= \operatorname{softmax}\left(\frac{\mathbf{Q}_h\mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{M}_i^{(t)} \right), \quad \text{head}_h = \boldsymbol{\alpha}_h\mathbf{U}_h, \\ \mathbf{H}_{i,\text{att}} &= \mathbf{H}_i^{(0)} + \operatorname{Dropout}_p\left([\text{head}_1 \,\|\, \cdots \,\|\, \text{head}_H]\mathbf{W}^O\right). \end{aligned} \tag{18}$$

Here, $\mathbf{W}_h^Q, \mathbf{W}_h^K, \mathbf{W}_h^U \in \mathbb{R}^{d_e \times d_h}$ and $\mathbf{W}^O \in \mathbb{R}^{d_e \times d_e}$, so every matrix dimension is explicit. $M_{i,rs}^{(t)} = -\infty$ blocks information from an unavailable sensor channel and is zero otherwise; when all channels are observed, $\mathbf{M}_i^{(t)} = \mathbf{0}$. Thus, attention learns which physical quantities should interact without erasing their channel identities.

The attended tokens are vectorized and projected to $\mathbf{h}_i^{(1)} \in \mathbb{R}^d$. Each of the following L residual blocks is written compactly as

$$\begin{aligned} \mathbf{u}_i^{(\ell)} &= \operatorname{ReLU}\left(\operatorname{LN}_1^{(\ell)}(\mathbf{h}_i^{(\ell)}\mathbf{W}_1^{(\ell)} + \mathbf{b}_1^{(\ell)}) \right), \\ \mathcal{R}_\ell(\mathbf{h}_i^{(\ell)}) &= \operatorname{LN}_2^{(\ell)}\left(\mathbf{u}_i^{(\ell)}\mathbf{W}_2^{(\ell)} + \mathbf{b}_2^{(\ell)} \right), \\ \mathbf{h}_i^{(\ell+1)} &= \operatorname{ReLU}\left(\mathbf{h}_i^{(\ell)} + \mathcal{R}_\ell(\mathbf{h}_i^{(\ell)}) \right). \end{aligned} \tag{19}$$

The identity shortcut preserves primitive kinematic information and provides a direct gradient path. It improves optimization conditioning, but no strict nonvanishing-gradient lower bound is claimed because nonlinear branch derivatives can cancel and ReLU units can be inactive.

A defender-specific GRU state $\mathbf{g}_i^{(t)}$ summarizes the engagement history, after which the shared actor samples a bounded continuous guidance action:

$$\begin{aligned} \mathbf{g}_i^{(t)} &= \operatorname{GRU}(\mathbf{h}_i^{(L+1)}, \mathbf{g}_i^{(t-1)}; \mathbf{W}_{\text{GRU}}), \\ \mathbf{a}_i^{(t)} &\sim \pi_\theta(\cdot \mid \mathbf{o}_i^{(t)}) = \mathcal{N}\left(\boldsymbol{\mu}_\theta(\mathbf{g}_i^{(t)}), \operatorname{diag}(\boldsymbol{\sigma}_\theta^2(\mathbf{g}_i^{(t)})) \right), \\ \mathbf{a}_i^{(t)} &= [n_{x_i}^{(t)}, n_{y_i}^{(t)}, n_{z_i}^{(t)}]^\top \in \mathcal{A}. \end{aligned} \tag{20}$$

The sampled action is clipped to the three-dimensional load-factor envelope $\mathcal{A} = [-0.1, 1] \times [-1, 1] \times [-1, 1]$. The three components regulate speed, heading, and flight-path angle through Eq. (**??**), respectively.

During training, the centralized critic receives the joint state

$$\mathbf{s}_t = [\mathbf{o}_1^{(t)} \| \cdots \| \mathbf{o}_{N_D}^{(t)}] \in \mathbb{R}^{N_D n_o}, \qquad V_\phi(\mathbf{s}_t) \in \mathbb{R}. \tag{21}$$

At deployment, each actor uses only its own observation and recurrent state.

B. Trust-Modulated Tactical Exploration

Trust is a training-time memory of how reliably a defender performs relative to the recent swarm baseline; it is not a flight-safety certificate. Let $G_i^{(k)} = \sum_{t=0}^{H_{\text{ep}}-1} \gamma_{\text{RL}}^t r_i^{(t)}$ be the return of defender i in episode k. Batch statistics and their exponential moving averages are

$$\begin{aligned} &\bar{G}^{(k)} = \frac{1}{N_D}\sum_{i=1}^{N_D} G_i^{(k)}, \qquad v_G^{(k)} = \frac{1}{N_D}\sum_{i=1}^{N_D}(G_i^{(k)} - \bar{G}^{(k)})^2, \\ &\mu_G \leftarrow (1-\rho)\mu_G + \rho\bar{G}^{(k)}, \qquad \sigma_G^2 \leftarrow (1-\rho)\sigma_G^2 + \rho v_G^{(k)}, \\ &\widetilde{G}_i^{(k)} = \frac{G_i^{(k)} - \mu_G}{\sqrt{\sigma_G^2 + \varepsilon_{\text{num}}}}. \end{aligned} \tag{22}$$

The trust state is then updated by

$$\mathcal{T}_i^{(k+1)} = (1-\alpha_T)\mathcal{T}_i^{(k)} + \alpha_T\sigma(\tau_T\widetilde{G}_i^{(k)}), \tag{23}$$

where $\alpha_T \in (0,1)$ controls memory and $\tau_T > 0$ controls sensitivity. Above-baseline returns increase trust and reduce additional exploration; below-baseline returns do the opposite.

During training, the action distribution is a trust-dependent mixture

$$\begin{aligned} \pi_i^{\text{mix}}(\mathbf{a} \mid \mathbf{o}_i) &= (1-\beta_i)\pi_\theta(\mathbf{a} \mid \mathbf{o}_i) + \beta_i\pi_i^{\text{guide}}(\mathbf{a}), \\ \pi_i^{\text{guide}}(\mathbf{a}) &= \omega_{\text{PN}}\delta(\mathbf{a} - \mathbf{a}_{\text{PN}}) + \omega_{\text{pr}}\delta(\mathbf{a} - \mathbf{a}_{\text{probe}}) \\ &\quad + \omega_{\text{rnd}}\mathcal{U}(\mathcal{A}), \\ \beta_i &= 1 - \mathcal{T}_i, \qquad \omega_{\text{PN}} + \omega_{\text{pr}} + \omega_{\text{rnd}} = 1. \end{aligned} \tag{24}$$

$\mathbf{a}_{\text{PN}}$ is the classical PN command clipped to $\mathcal{A}$. $\mathbf{a}_{\text{probe}}$ is a feasible boundary command whose sign is chosen to reduce the current velocity-to-LOS error; it probes maneuver authority without invoking an undefined action-value function or intentionally selecting the worst-valued action. The uniform term maintains residual coverage. At deployment, guided exploration is disabled ($\beta_i = 0$), so the decentralized learned policy supplies the command.

C. Dual-Clip Optimization with Adaptive KL Regularization

For a rollout, generalized advantage estimation is computed from

$$\delta_t = r_i^{(t)} + \gamma_{\mathrm{RL}} V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t), \quad \widehat{A}_t = \sum_{\ell=0}^{H_{\mathrm{ep}}-t-1} (\gamma_{\mathrm{RL}}\lambda_{\mathrm{GAE}})^\ell \delta_{t+\ell}. \tag{25}$$

Let $\varrho_t(\theta) = \pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t)/\pi_{\theta_{\mathrm{old}}}(\mathbf{a}_t \mid \mathbf{o}_t)$ denote the importance ratio. The standard and dual-clipped surrogates are combined in one definition:

$$\begin{aligned} \bar{\varrho}_t &= \mathrm{clip}(\varrho_t, 1-\varepsilon_{\mathrm{PPO}}, 1+\varepsilon_{\mathrm{PPO}}), \\ L_t^{\mathrm{clip}} &= \min(\varrho_t \widehat{A}_t, \bar{\varrho}_t \widehat{A}_t), \\ J^{\mathrm{DC}}(\theta) &= \mathbb{E}_t\left[\begin{cases} \max(L_t^{\mathrm{clip}}, c_{\mathrm{DC}}\widehat{A}_t), & \widehat{A}_t < 0, \\ L_t^{\mathrm{clip}}, & \widehat{A}_t \geq 0, \end{cases}\right]. \end{aligned} \tag{26}$$

For negative-advantage samples, the additional floor prevents an arbitrarily unfavorable surrogate contribution. It clips the optimization objective; it does not impose a hard two-sided interval on the realized probability ratio. We use $c_{\mathrm{DC}} > 1 + \varepsilon_{\mathrm{PPO}}$.

The empirical policy divergence is $\widehat{D}_{\mathrm{KL}} = \mathbb{E}_t[\log \pi_{\theta_{\mathrm{old}}}(\mathbf{a}_t \mid \mathbf{o}_t) - \log \pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t)]$. Its penalty coefficient adapts after epoch k as

$$\beta_{\mathrm{KL}}^{(k+1)} = \begin{cases} \min(1.5\beta_{\mathrm{KL}}^{(k)}, \beta_{\max}), & \widehat{D}_{\mathrm{KL}} > 2\delta_{\mathrm{KL}}, \\ \max(0.9\beta_{\mathrm{KL}}^{(k)}, \beta_{\min}), & \widehat{D}_{\mathrm{KL}} < 0.5\delta_{\mathrm{KL}}, \\ \beta_{\mathrm{KL}}^{(k)}, & \text{otherwise}. \end{cases} \tag{27}$$

Thus, excessive policy drift strengthens the next update's penalty, while very small drift relaxes it. This mechanism controls update speed; any relationship to load-factor smoothness is validated empirically rather than treated as a structural load bound.

With $\widehat{V}_t^{\mathrm{targ}} = \widehat{A}_t + V_\phi(\mathbf{s}_t)$, define $L_V = B^{-1}\sum_{t=1}^B (V_\phi(\mathbf{s}_t) - \widehat{V}_t^{\mathrm{targ}})^2$ and $S[\pi_\theta] = -\mathbb{E}_t[\log \pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t)]$. The complete training loss is

$$\mathcal{L}(\theta, \phi) = -J^{\mathrm{DC}}(\theta) + \beta_{\mathrm{KL}}\widehat{D}_{\mathrm{KL}} + c_v L_V - c_e S[\pi_\theta]. \tag{28}$$

The actor minimizes the policy, KL, and entropy terms; the critic minimizes L_V. AdamW with the cosine schedule reported in Table **??** performs the updates.

D. State Representation and Reward Design

The local observation retains relative three-dimensional geometry, cooperative timing, and the previous applied command. Using fixed physical scales, it is

$$\begin{aligned} \mathbf{o}_i^{(t)} = \mathrm{clip}\big([&d_i/d_{\mathrm{ref}},\ \Delta z_i/z_{\mathrm{ref}},\ \dot{d}_i/V_{\mathrm{ref}},\ e_{\angle,i}/\pi, \\ &V_i/V_{\mathrm{ref}},\ V_{T(i)}/V_{\mathrm{ref}},\ \theta_i/\pi,\ \gamma_i/\pi,\ \boldsymbol{\ell}_{iT}^\top, \\ &(t_{go,i} - \bar{t}_{go,i})/T_{\mathrm{ref}},\ (t_{go,i}^{\max} - t_{go,i}^{\min})/T_{\mathrm{ref}}, \\ &(t_{go,i} - t_{go,i}^{\min})/T_{\mathrm{ref}},\ (\mathbf{a}_i^{(t-1)})^\top]^\top, -1, 1\big) \in \mathbb{R}^{17}. \end{aligned} \tag{29}$$

Here, Δz_i is the defender–target altitude difference, $\boldsymbol{\ell}_{iT} \in \mathbb{R}^3$ is the LOS unit vector, and $\hat{\mathbf{v}}_i = \mathbf{v}_i/(\|\mathbf{v}_i\|_2 + \varepsilon_{\mathrm{num}})$ is the unit velocity direction. Therefore, $e_{\angle,i} = \arccos[\mathrm{clip}(\hat{\mathbf{v}}_i^\top \boldsymbol{\ell}_{iT}, -1, 1)]$ is the three-dimensional velocity-to-LOS angle. The mean, minimum, and maximum time-to-go values are computed over the cooperative group $\mathcal{G}_i$. The previous command is available before the current action is selected and supplies short-term actuator history without violating causality. Fixed scales $d_{\mathrm{ref}}, z_{\mathrm{ref}}, V_{\mathrm{ref}}$, and T_{ref} make the heterogeneous physical channels comparable.

TABLE I
REWARD COMPONENTS AND PHYSICAL ROLES

Component	Definition and role
r_i^{dist}	$\exp[-\|\mathbf{p}_i - \mathbf{p}_{\mathrm{tgt}}\|/d_{\mathrm{ref}}]$; bounded dense feedback for closing range.
r_i^{angle}	$-e_{\angle,i}$; penalizes three-dimensional velocity-to-LOS misalignment.
r_i^{hit}	$R_{\mathrm{hit}}\mathbb{I}[\text{first entry into } d_{\mathrm{kill}}]$; one-time terminal success reward.
r_i^{coord}	$-\lvert t_{go,i} - \lvert\mathcal{G}_i\rvert^{-1}\sum_{j\in\mathcal{G}_i} t_{go,j}\rvert$; promotes simultaneous arrival.
r_i^{effort}	$-[\|\mathbf{a}_i^{(t)}\|_2^2 + \kappa_\Delta\|\mathbf{a}_i^{(t)} - \mathbf{a}_i^{(t-1)}\|_2^2]$; penalizes all three load channels and their command variation.

The reward connects each physical objective to one interpretable component:

$$r_i^{(t)} = \lambda_d r_i^{\mathrm{dist}} + \lambda_q r_i^{\mathrm{angle}} + \lambda_h r_i^{\mathrm{hit}} + \lambda_c r_i^{\mathrm{coord}} + \lambda_u r_i^{\mathrm{effort}}. \tag{30}$$

The components are summarized below to avoid another rapid sequence of five displayed equations. The environment computes these rewards during training; the critic estimates their discounted return but does not generate the reward. At deployment, no reward or centralized critic is required.

E. Sensitivity of Reward Weights and Hyperparameters

The weights λ_d and λ_q regulate early approach and heading alignment, whereas λ_h and λ_c determine the balance between terminal interception and coordinated arrival. Excessive coordination weight can delay a defender with favorable geometry; excessive effort weight λ_u can suppress the normal-load commands required against a maneuvering target. We therefore normalize all reward components first, tune hit/coordination weights around mission success, and then increase the effort term until command peaks and variations satisfy the desired envelope without degrading interception reliability.

The PPO clipping factor $\varepsilon_{\mathrm{PPO}}$, dual-clip parameter c_{DC}, target divergence δ_{KL}, and β_{KL} adaptation determine policy update conservatism. Smaller clipping/divergence targets improve stability but can slow adaptation; larger values learn aggressive maneuvers faster but increase return variance. λ_{GAE} and c_e control the bias–variance and exploration–exploitation tradeoffs. Their numerical settings and sensitivity tests are reported with the simulation results rather than expanded into additional equations here.

V. Simulation Verification

A. Experimental Setup and Implementation Details

All training and evaluation were performed on an Intel Core i7-9700F CPU / Nvidia RTX 3090 GPU server running rlsuanfa.pdf

Fig. 5. ART-MAPPO information flow from physical observations through attention-residual and recurrent encoding to trust-modulated exploration and regularized actor–critic updates.

Ubuntu 18.04. The simulation step size was 0.05 s and the maximum episode duration was 75 s. Table **??** summarizes the key hyperparameters of ART-MAPPO.

The horizontal projection of the simulation environment is depicted in Fig. **??**. In the three-dimensional case presets, the 20 defenders start near the defended center at radii 52.63–65.44 m and altitude 0 m, while the eight attackers start at radii 1987.72–2145.95 m and altitude 120 m. Initial defender and attacker speeds are 20.32–30.02 m/s and 30.23–45.87 m/s, respectively. The attackers initially point toward the defended region and then execute the case-specific three-dimensional maneuvers described below.

A representative scenario of 20 D_{UAV} versus 8 A_{UAV} is considered. Table **??** summarizes the audited three-dimensional initialization and kinematic constraints.

Two dynamic engagement scenarios are designed to evaluate cooperative guidance performance. The scenario configurations are listed in Table **??**.

To represent delayed communication and sensing uncertainty, every defender receives the position and velocity packets used to construct Eq. (**??**) after an end-to-end delay of 50 ms. For each Cartesian component, $\widehat{\mathbf{p}}(t) = \mathbf{p}(t - 50\,\mathrm{ms}) +$

Algorithm 2 ART-MAPPO Cooperative Interception Guidance

Require: Consensus assignment $\mathbf{X}^*$, environment, episode horizon H_{ep}, actor θ, critic ϕ, trust and PPO parameters
Ensure: Decentralized shared guidance policy π_θ

```
Initialize networks, GRU states g_i, trust states T_i, and buffer D
for episode k = 1, 2, ... do
    Reset the scenario and form cooperative groups from X*
    for t = 0 to H_ep − 1 do
        for each defender i do
            Construct o_i^(t) using (??)
            Encode attention/residual features and update g_i^(t)
            Sample a_i^(t) from (??) and clip it to A
        end for
        Execute the joint action and compute (??)
        Store the transition in D
    end for
    Update return statistics and trust using (??)–(??)
    Estimate Â_t using (??)
    Update actor and critic by minimizing (??); adapt β_KL using (??)
    Clear D
end for
Deploy π_θ with local observations, local GRU states, and β_i = 0
```

TABLE II
ART-MAPPO ALGORITHM PARAMETERS

Parameter	Value
PPO clipping factor (ε_{PPO})	0.2
Dual-clip parameter (c_{DC})	3.0
Target KL divergence (δ_{KL})	0.02
Entropy bonus coefficient (c_e)	0.01
GAE parameter (λ_{GAE})	0.97
LR schedule	Cosine Annealing
Optimizer	AdamW
Mini-batch number	4
Buffer size	4096
Learning rate	3×10^{-4}
Attention heads (H)	4
Residual blocks (L)	2

$\boldsymbol{\eta}_p$ with $\boldsymbol{\eta}_p \sim \mathcal{N}(\mathbf{0}\,\mathrm{m}, 9\mathbf{I}_3\,\mathrm{m}^2)$, and $\widehat{\mathbf{v}}(t) = \mathbf{v}(t - 50\,\mathrm{ms}) + \boldsymbol{\eta}_v$ with $\boldsymbol{\eta}_v \sim \mathcal{N}(\mathbf{0}\,\mathrm{m/s}, 0.09\mathbf{I}_3\,\mathrm{m}^2/\mathrm{s}^2)$. Thus, every position component has mean error $0\,\mathrm{m}$ and variance $9\,\mathrm{m}^2$, while every velocity component has mean error $0\,\mathrm{m/s}$ and variance $0.09\,(\mathrm{m/s})^2$. Range, altitude difference, LOS direction, closing speed, heading, pitch, velocity-to-LOS angle, and time-to-go are recomputed from these delayed noisy Cartesian states; no additional independent noise is imposed on those derived quantities. Previous applied loads are internally measured without additive observation noise, while the commanded loads pass through a first-order response with a 300-ms time constant. Table **??** summarizes these settings in physical units.

B. Evaluation Criteria

The evaluation metrics are defined from quantities measured at interception, rather than from time-to-go values that vanish at the terminal instant. For target t_j, let $\mathcal{G}_j = \{i : X_{ij} = 1\}$ be its cooperative defender group, $m_j = |\mathcal{G}_j|$, t_i^{arr} the first time defender i reaches the lethal radius, and $\mathcal{W}_i = [t_i^{\text{arr}} - \Delta T_{\text{term}}, t_i^{\text{arr}}]$ the terminal measurement window. The four group-level metrics are collected in one equation:

$$
\begin{aligned}
\bar{t}_j^{\text{arr}} &= \frac{1}{m_j} \sum_{i \in \mathcal{G}_j} t_i^{\text{arr}}, \\
E_{\text{co-time}}^{(j)} &= \frac{1}{m_j} \sum_{i \in \mathcal{G}_j} \left| t_i^{\text{arr}} - \bar{t}_j^{\text{arr}} \right|, \\
\bar{n}_{i,\text{term}} &= \frac{1}{|\mathcal{W}_i|} \sum_{t \in \mathcal{W}_i} \sqrt{(n_{y_i}^{(t)})^2 + (n_{z_i}^{(t)})^2}, \\
E_n^{(j)} &= \frac{1}{m_j} \sum_{i \in \mathcal{G}_j} \bar{n}_{i,\text{term}}, \\
E_{\text{miss}}^{(j)} &= \frac{1}{m_j} \sum_{i \in \mathcal{G}_j} d_i(t_i^{\text{arr}}), \\
E_t^{(j)} &= \max_{i \in \mathcal{G}_j} t_i^{\text{arr}} - t_0.
\end{aligned}
\tag{31}
$$

Here, $E_{\text{co-time}}$ measures arrival-time dispersion, E_n is the mean terminal-window resultant normal load factor, E_{miss} is the terminal miss distance, and E_t is the time from engagement start t_0 until the last member of the group arrives. This separates synchronization quality from total engagement duration. A trial-level value is the mean of the corresponding group values over the eight targets. Monte Carlo boxplots use successful trials only; interception success rate is reported separately so that failed trials are not hidden by the conditional metric distributions.

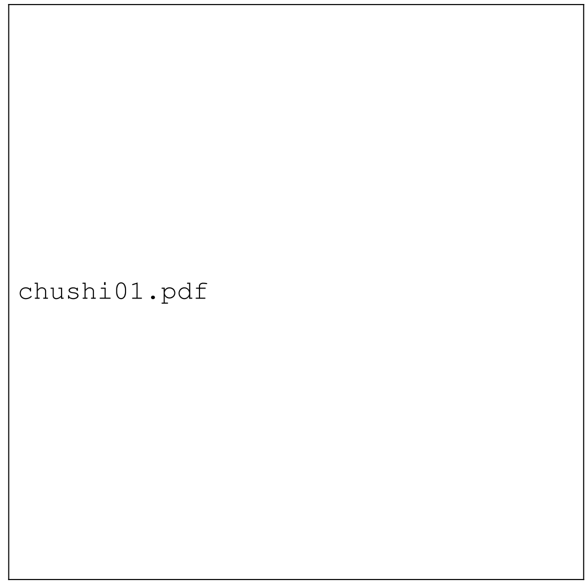


Fig. 6. Simulation environment configuration for training.

C. Target Assignment Performance and Spatial Topology

To evaluate the IDBO algorithm, it is benchmarked against PSO, GWO, SSA, BOA, and standard DBO.

Fig. **??** illustrates the statistical performance in terms of average cost convergence over iterations. IDBO significantly outperforms the baselines, reaching the lowest steady-state cost within 100 iterations.

TABLE III
UAV PARAMETERS FOR ATTACKERS AND DEFENDERS

	Initial radius / altitude	Initial velocity V	Initial heading γ	Maneuverability (n_x, n_y, n_z)
D_{UAV}	52.63–65.44 m / 0 m	20.32–30.02 m/s	$\gamma_D \in [0, 2\pi]$	$n_x \in [-0.1, 1]$, $n_y, n_z \in [-1, 1]$
A_{UAV}	1987.72–2145.95 m / 120 m	30.23–45.87 m/s	Toward defended area	Case-specific 3-D maneuver

TABLE IV
COMBAT SCENARIO SETTINGS

	$N_{D\text{-UAV}}$	$N_{A\text{-UAV}}$	A_{UAV} Strategy
Case 1	20	8	Hybrid Evasion
Case 2	20	8	Sinusoidal Maneuver

TABLE V
COMMUNICATION DELAY AND OBSERVATION-UNCERTAINTY SETTINGS

Observed quantity	Delay / response	Error distribution
Position x, y, z	50 ms	Mean 0 m; variance 9 m^2
Velocity v_x, v_y, v_z	50 ms	Mean 0 m/s; variance 0.09 (m/s)^2
Derived geometry and t_{go}	50 ms	No independent noise; computed from $\widehat{\mathbf{p}}, \widehat{\mathbf{v}}$
Previous loads n_x, n_y, n_z	300 ms response	Mean 0; variance 0

convergence.png

Fig. 7. Average cost convergence over iterations for target assignment algorithms.

Furthermore, the violin plot in Fig. **??** evaluates the algorithmic robustness across multiple independent runs. It shows IDBO's compact distribution at the lowest median cost, highlighting its superior stability and consistency compared to the other heuristic methods.

The optimal assignment mapping generated by the algorithm is visualized in Fig. **??**. Attackers A_1–A_4 are each allocated three defenders, while A_5–A_8 are intercepted by two defenders each, perfectly respecting the per-target capacity constraint $L_{\max}$ and ensuring sufficient numerical superiority for cooperative strikes.

violin.png

Fig. 8. Violin plot of total cost distribution evaluating algorithm robustness.

target_assignment_topology.png

Fig. 9. Visualized spatial topology of the target assignment results.

D. RL Model Training and Convergence Analysis

To evaluate the learning efficiency and robustness of the proposed ART-MAPPO, comparative training experiments were conducted against standard MAPPO, IPPO, IA2C, and IQL. All models were trained over 10,000 episodes with a maximum horizon of 25 steps per episode. To ensure statistical significance and eliminate the bias of favorable network initialization, all learning curves in Fig. **??** are averaged across 5 independent random seeds, with the shaded regions denoting the standard deviation.

As illustrated in Fig. **??**(a), ART-MAPPO demonstrates a superior learning paradigm, achieving the highest asymptotic normalized reward of 8.97 ± 0.64 (averaged over the final 500 episodes). This represents a notable performance improvement of 8.2%, 12.3%, 14.4%, and 29.8% over IQL, MAPPO, IPPO, and IA2C, respectively. The advantage of ART-MAPPO is particularly pronounced during the intermediate training phase. By episode 4,000, ART-MAPPO establishes a significant performance leap, outperforming standard MAPPO by 43.6% and IQL by 16.6%, highlighting its ability to rapidly escape local optima early in the training process.

Furthermore, the proposed method exhibits accelerated convergence alongside enhanced training stability. When evaluated on a 100-episode smoothed curve, ART-MAPPO reaches 90% of its optimal asymptotic performance by episode 5,591. In contrast, the most competitive baseline, IQL, requires 6,477 episodes (a 13.7% delay), while standard MAPPO lags significantly, requiring 7,654 episodes. Regarding algorithmic stability, ART-MAPPO maintains a tight variance across random seeds ($\sigma = 0.64$), whereas standard MAPPO exhibits a standard deviation of 1.34 (approximately twice as large). This confirms that the proposed enhancements effectively mitigate the high-variance gradient updates typical in multi-agent policy-based methods.

These macroscopic performance gains are directly supported by the network-level metrics. The critic loss (Fig. **??**(b)) of ART-MAPPO converges to the lowest steady-state error with minimal oscillation, a direct consequence of the dual-clip mechanism and adaptive KL penalty. Concurrently, the policy entropy (Fig. **??**(c)) demonstrates an optimal, gradual decay trajectory, validating that the trust-modulated exploration mechanism successfully balances early-stage extensive state-space exploration with late-stage strict policy exploitation.

It should be emphasized that the convergence discussed here is empirical training convergence rather than a global optimality guarantee for the nonconvex MARL problem. In particular, ART-MAPPO inherits the practical convergence interpretation of PPO-style policy gradient methods: a stable learning process is indicated by a saturating reward curve, bounded inter-seed variance, a nondivergent critic loss, and a gradually decreasing policy entropy. The dual-clip objective and adaptive KL penalty do not prove convergence to the global optimum; instead, they restrict the policy update magnitude and reduce destructive policy oscillations caused by large importance-sampling ratios. Therefore, the convergence claim for ART-MAPPO should be understood as statistically validated stability and asymptotic performance under the tested training settings, supported by the five-seed learning curves and the network-level diagnostics in Fig. **??**.

E. Offline Reward-Function Sensitivity Analysis

We further examine the sensitivity of the reward definition to three representative weights and one nonlinear shaping hyperparameter. The implementation keys w_{hit}, w_{coord}, and w_{energy} correspond to the terminal, coordination, and effort weights λ_h, λ_c, and λ_u in (**??**), respectively, while $\alpha_d \equiv d_{\text{ref}}^{-1}$ determines the exponential distance-reward decay. Their nominal values are $w_{\text{hit}} = 200$, $w_{\text{coord}} = 18$, $w_{\text{energy}} = 0.005$, and $\alpha_d = 2 \times 10^{-4}\,\text{m}^{-1}$. Each parameter is independently multiplied by $\kappa \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ while all other reward parameters remain nominal.

This experiment isolates reward-function sensitivity from policy adaptation. Specifically, the trained nominal ART-MAPPO policies are frozen, and no gradient, optimizer step, or policy update is performed. We collect 100 stochastic evaluation episodes (50 per case) and store the distance, time-to-go mismatch, three-axis control effort, and first-hit events. In every recorded episode, all 20 defenders reach their assigned targets and all eight attackers are intercepted, so the sparse hit contribution is evaluated on successful trajectories without changing any event indicator. Every parameter setting is then evaluated by recomputing the reward on these *same* state–action trajectories. For episode e, the normalized cumulative-reward variation is

$$\Delta G_e^{\text{norm}}(\kappa;\theta) = \frac{G_e(\kappa\theta_0) - G_e(\theta_0)}{\sum_t |r_e(t;\theta_0)| + 10^{-8}}, \tag{32}$$

where $G_e(\theta) = \sum_t r_e(t;\theta)$. Thus, Fig. **??** characterizes the mapping $\theta \mapsto r(s,a;\theta)$ for fixed trajectories rather than the performance of policies retrained under altered rewards.

The component-level responses in Figs. **??**(a)–(c) agree with the reward definition. Reducing α_d to $1.0 \times 10^{-4}\,\text{m}^{-1}$ extends the distance-reward e-folding range from the nominal 5.0 km to 10.0 km, whereas increasing it to $3.0 \times 10^{-4}\,\text{m}^{-1}$ contracts that range to 3.33 km. The resulting separation of the curved profiles over the observed 0–2.063 km range confirms a nonlinear, rather than merely amplitude-scaled, change in the reward landscape. In contrast, the coordination and effort terms change linearly with their outer weights. At the nominal setting, the coordination slope is $-w_{\text{coord}}\alpha_c = -2.88$ reward units per second of time-to-go mismatch; the 0.5–1.5× sweep changes this slope from -1.44 to -4.32. Similarly, the effort-penalty slope changes from -2.5×10^{-5} to -7.5×10^{-5} per unit of $n_x^2 + n_y^2 + n_z^2$.

For a dimensionless comparison around the nominal point, we use

$$S_\theta = \frac{1}{100}\sum_{e=1}^{100} \frac{|\Delta G_e^{\text{norm}}(1.25;\theta) - \Delta G_e^{\text{norm}}(0.75;\theta)|}{0.5}. \tag{33}$$

The resulting indices are 0.995913 for w_{hit}, 0.001317 for w_{coord}, 0.000823 for α_d, and 1.35×10^{-8} for w_{energy}. The terminal weight is therefore the dominant cumulative-reward sensitivity for these successful fixed trajectories: the mean nominal hit contribution is 8.80×10^6, compared with a mean total return of 8.825×10^6. The coordination and distance-shaping parameters have substantially smaller but clearly resolved effects around the nominal point. In particular, increasing w_{coord} steepens the penalty for any given arrival-time mismatch, while α_d produces the expected asymmetric nonlinear response in Fig. **??**(d). The very small effort index reflects its small nominal coefficient and the bounded control effort on the evaluated trajectories; it does not imply that high control effort is unpenalized, because panel (c) still shows the proportional increase in the instantaneous penalty. Finally, the variation is exactly zero at $\kappa = 1$ for all four parameters, and all confidence intervals are computed over the same 100 episodes, confirming a controlled one-factor-at-a-time comparison.

F. Cooperative Interception against Maneuvering Targets

Based on the optimal assignment topology, we evaluate three-dimensional cooperative guidance in two maneuvering scenarios. ART-MAPPO and the capacity-matched PN reference use the same initial states, target assignment, hit radius, speed limits, load-factor envelope, and other scenario settings described in Section **??**. For each method and case, we report the three-dimensional trajectory, its horizontal projection, all three load commands and their associated attitude/speed responses, time-to-go histories, and terminal arrival-time differences.

1) Case 1: Interception against Evasive Maneuvering Targets: In Case 1, the attacking UAVs initially execute a constant-load evasive maneuver and transition to PN guidance after 15 s. This creates simultaneous horizontal and vertical closing demands while the defenders must preserve the assignment groups and coordinate their terminal arrival times.

1) ART-MAPPO: Fig. **??** shows the complete spatial engagement. Starting below the attackers, the defenders climb from the defended region and close on the eight elevated target trajectories. The horizontal projection in Fig. **??** preserves the original plan-view interpretation and shows that all 20 defenders converge on the targets specified by the assignment topology.

The corresponding command and state histories are separated into three physically related pairs in Figs. **??**–**??**. The horizontal normal command n_y and heading γ_D form the yaw-plane response, while n_z and pitch θ_D show the vertical response that is absent from a planar model. ART-MAPPO keeps n_y inside the ± 1 envelope and uses a substantially smaller vertical command for most of the engagement. The smooth pitch histories confirm gradual altitude capture. Meanwhile, n_x and V_D show how the policy regulates closing speed without decoupling temporal coordination from the two normal-load channels.

The t_{go} histories in Fig. **??** converge within the eight target groups despite the three-dimensional engagement geometry. The terminal summary in Fig. **??** reports a maximum within-group arrival spread of 0.25 s, below the 0.5-s synchronization criterion.

2) PN baseline: The capacity-matched PN reference is shown in Figs. **??** and **??**. PN also completes the selected engagement, but its purely reactive commands produce visibly different spatial curvature and control transients. The same command limits are imposed on both methods, so the comparison does not attribute physically unavailable load authority to the baseline.

Figures **??**–**??** give the corresponding yaw-plane, pitch-plane, and axial responses. The PN commands are clipped to $n_x \in [-0.1, 1]$ and $n_y, n_z \in [-1, 1]$, and the command histories show more direct use of the available bounds. The pitch response remains finite and the speed histories remain within the common vehicle envelope, providing a physically matched three-dimensional reference.

The PN t_{go} histories also converge in this selected run, as shown in Fig. **??**. Its maximum terminal group spread is 0.40 s (Fig. **??**), compared with 0.25 s for ART-MAPPO. Both satisfy the 0.5-s criterion, while the learned policy gives the tighter representative Case 1 synchronization.

fig_a_reward.png

(a) Training Reward

fig_b_critic_loss.png

(b) Critic Loss

fig_c_entropy.png

(c) Policy Entropy

Fig. 10. Comparative training curves of ART-MAPPO and baseline algorithms over 10,000 episodes. Solid lines and shaded regions represent the mean and standard deviation across 5 independent random seeds, respectively. Notably,

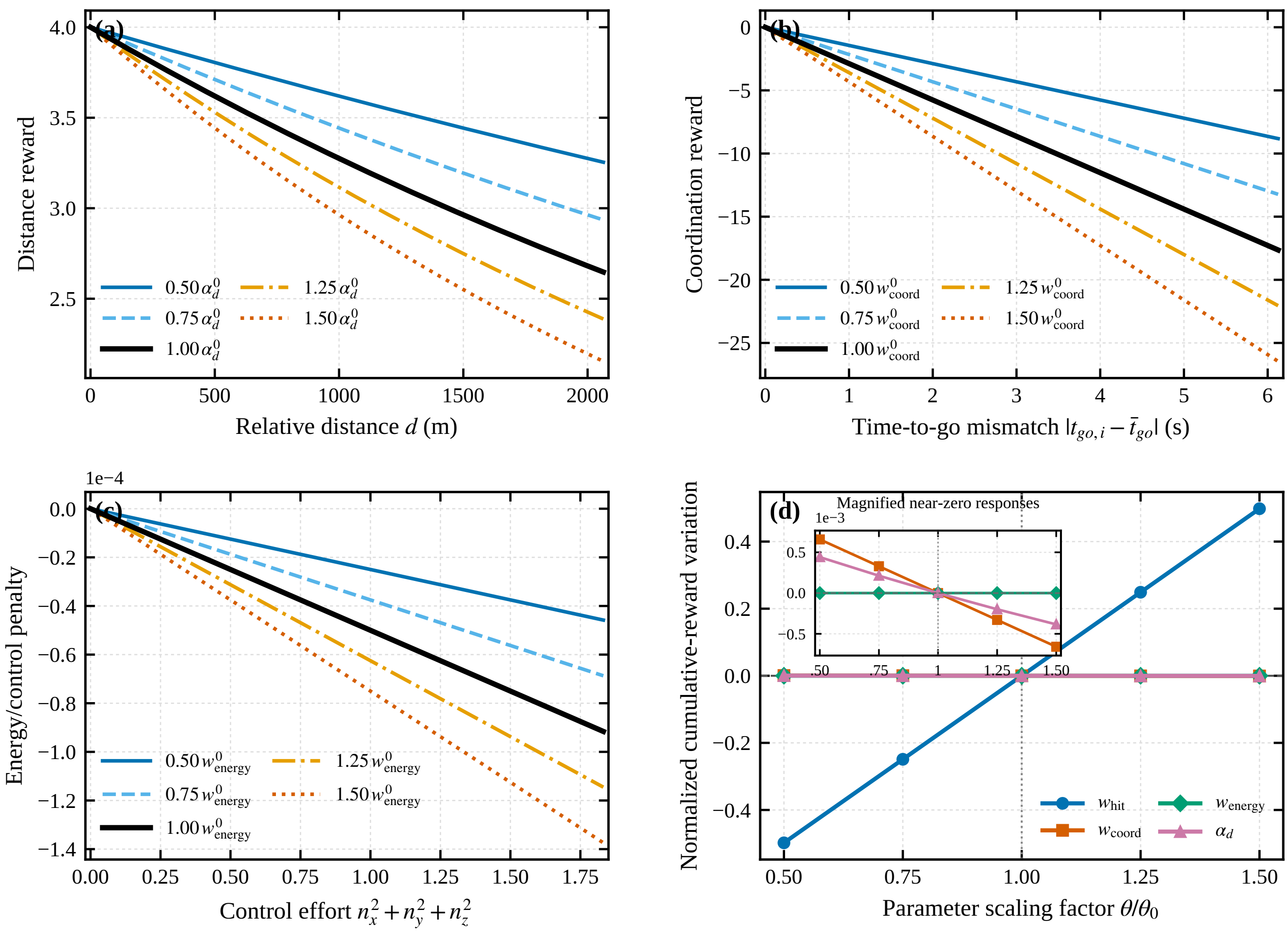


Fig. 11. Sensitivity of the reward function to representative weighting and shaping parameters. (a) Distance reward versus relative distance under variations of α_d. (b) Coordination reward versus time-to-go mismatch under variations of w_{coord}. (c) Energy/control penalty versus three-axis control effort under variations of w_{energy}. (d) Mean normalized cumulative-reward variation on 100 fixed ART-MAPPO evaluation trajectories; shaded regions denote 95% confidence intervals, and the inset magnifies the three near-zero responses.

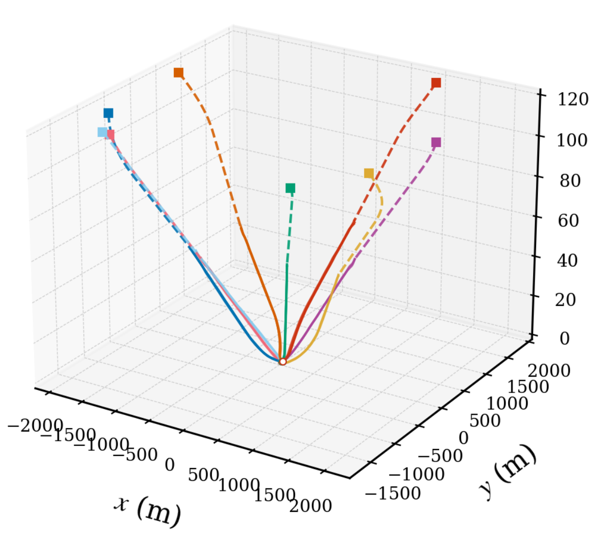


Fig. 12. Three-dimensional engagement trajectories of the defending swarm using ART-MAPPO in Case 1.

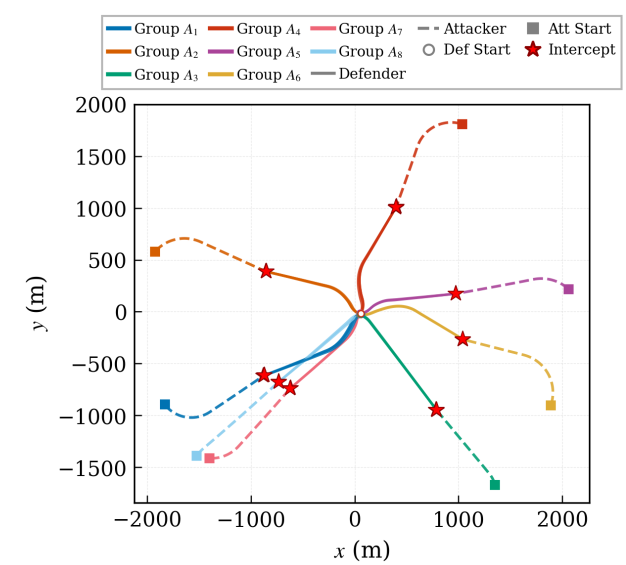
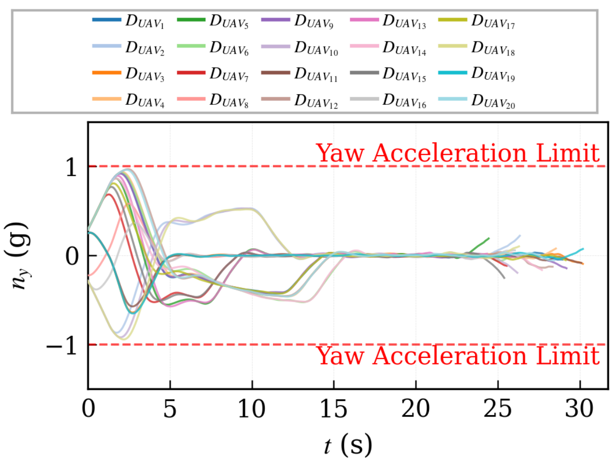
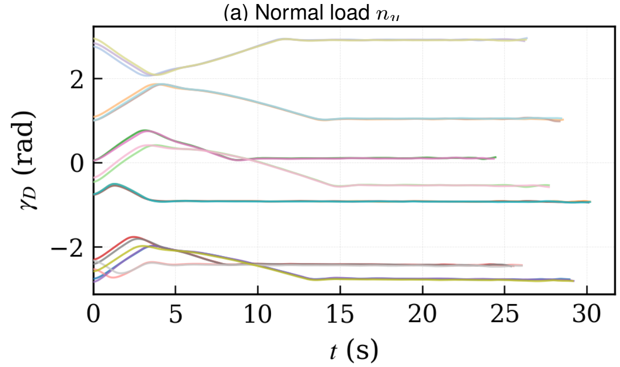


Fig. 13. Horizontal projection of the ART-MAPPO engagement trajectories in Case 1.

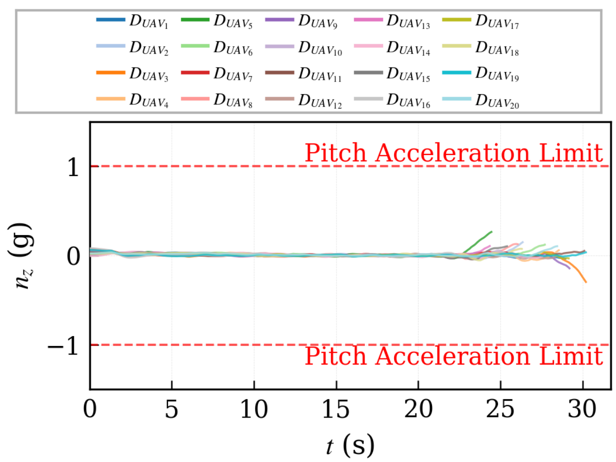


(a) Normal load n_y

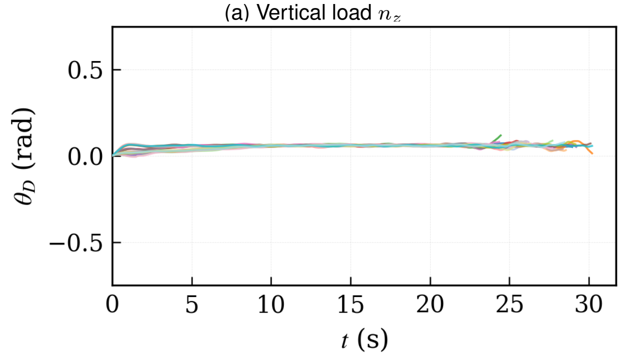


(b) Heading γ_D

Fig. 14. Yaw-plane command and response of ART-MAPPO in Case 1.

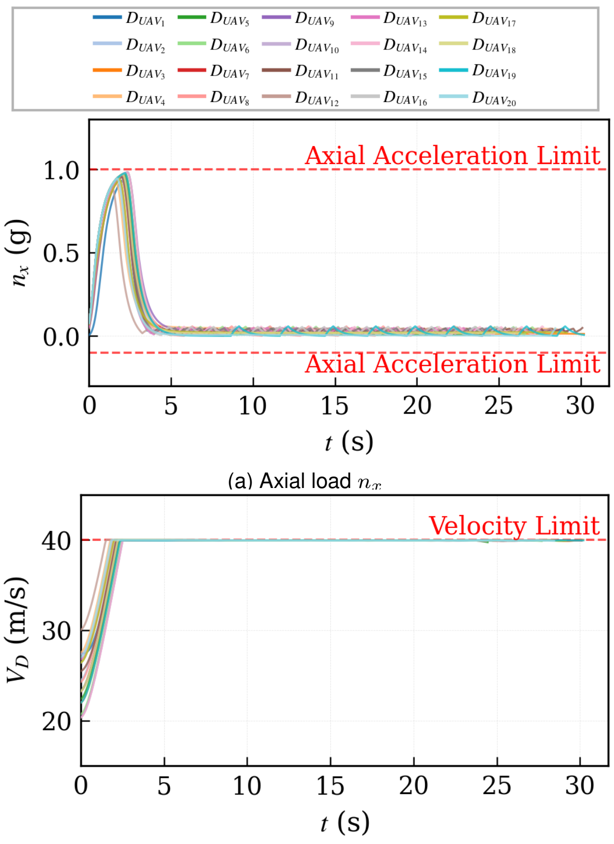


(a) Vertical load n_z



(b) Pitch θ_D

Fig. 15. Pitch-plane command and response of ART-MAPPO in Case 1.



(a) Axial load n_x

(b) Velocity V_D

Fig. 16. Axial command and speed response of ART-MAPPO in Case 1.

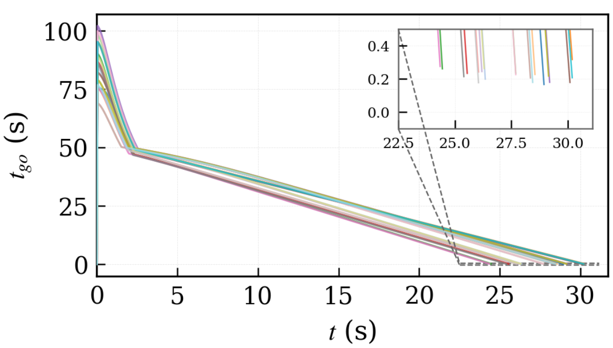


Fig. 17. Time-to-go (t_{go}) profiles of the defending swarm using ART-MAPPO in Case 1.

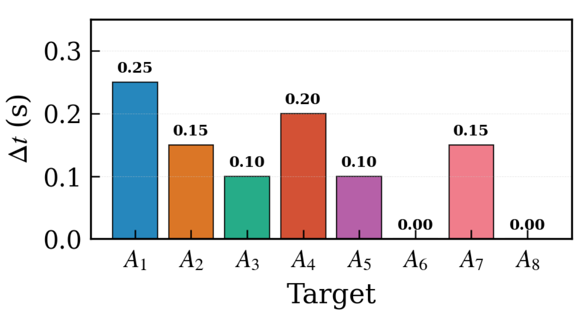


Fig. 18. Terminal arrival time differences Δt within each ART-MAPPO interceptor group in Case 1.

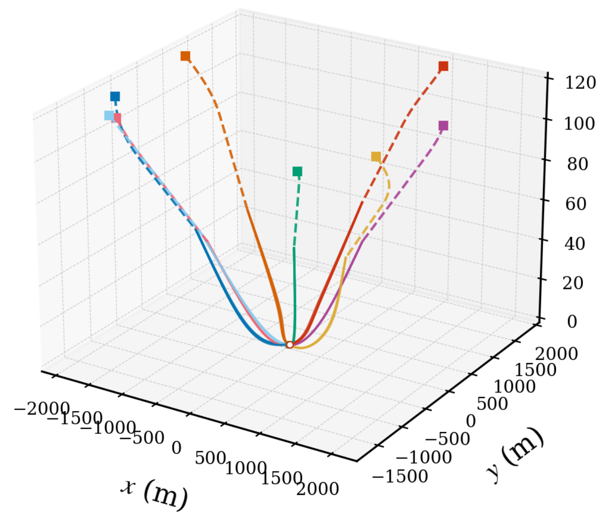


Fig. 19. Three-dimensional engagement trajectories of the defending swarm using the PN baseline in Case 1.

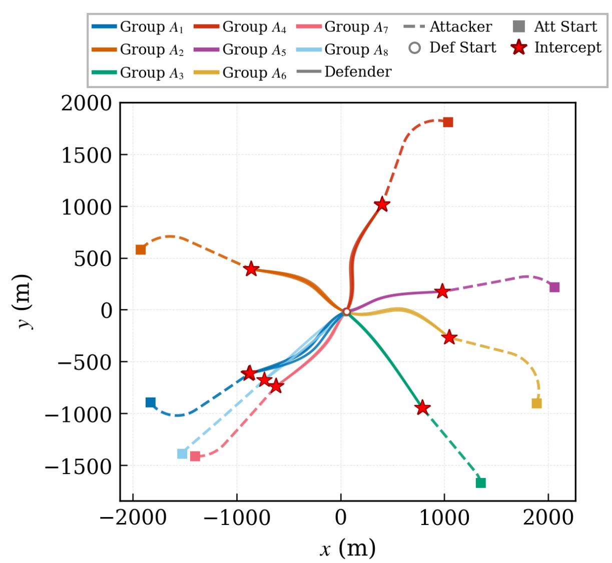
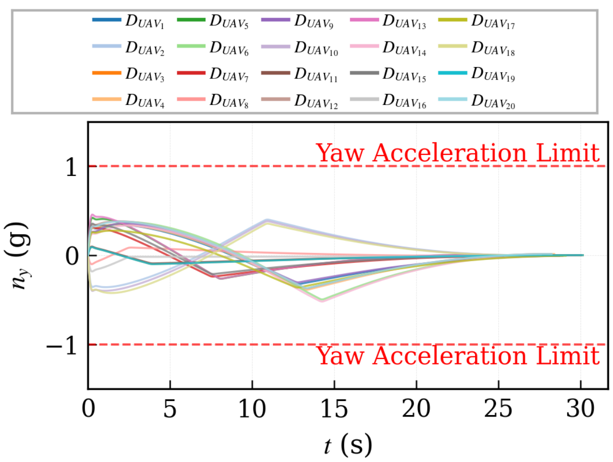
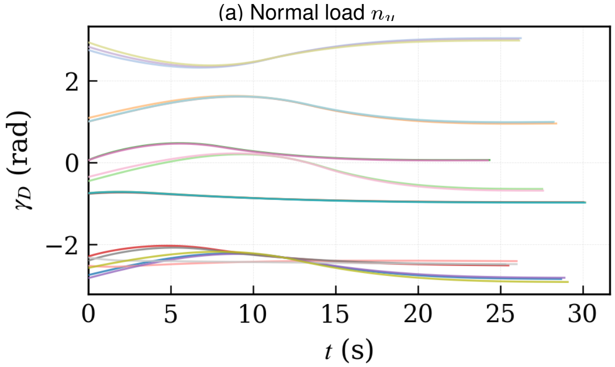


Fig. 20. Horizontal projection of the PN engagement trajectories in Case 1.

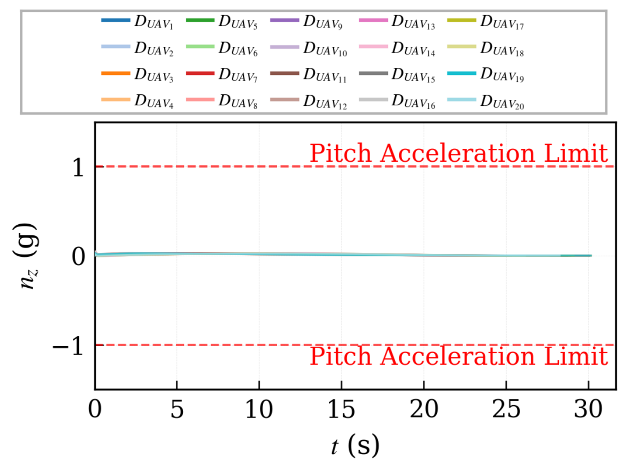


(a) Normal load n_y

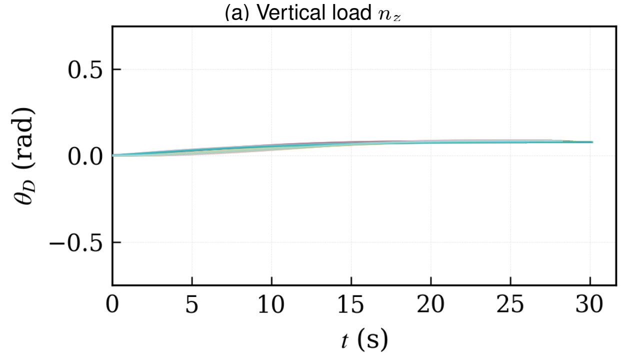


(b) Heading γ_D

Fig. 21. Yaw-plane command and response of the capacity-matched PN baseline in Case 1.

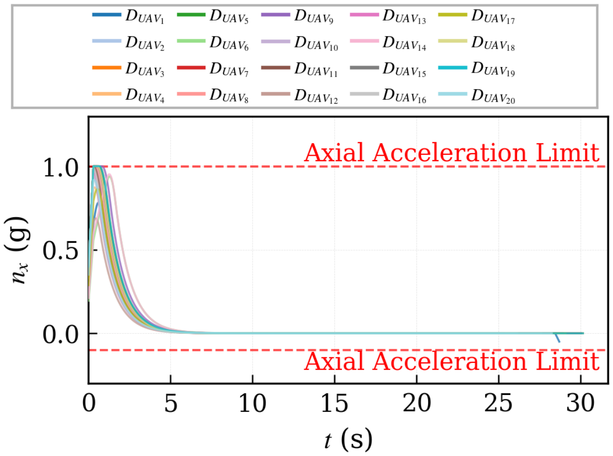


(a) Vertical load n_z

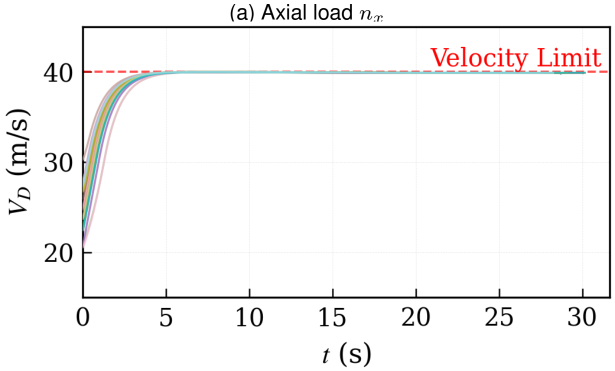


(b) Pitch θ_D

Fig. 22. Pitch-plane command and response of the capacity-matched PN baseline in Case 1.



(a) Axial load n_x



(b) Velocity V_D

Fig. 23. Axial command and speed response of the capacity-matched PN baseline in Case 1.

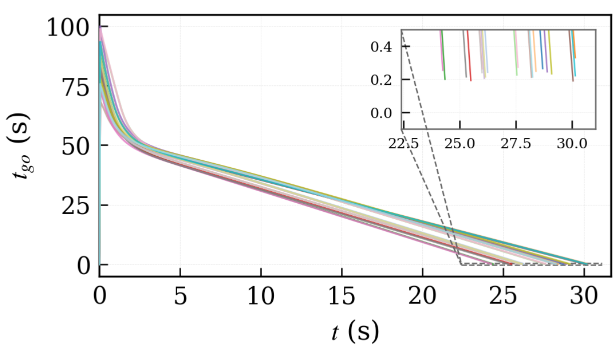


Fig. 24. Time-to-go (t_{go}) profiles of the defending swarm using the PN baseline in Case 1.

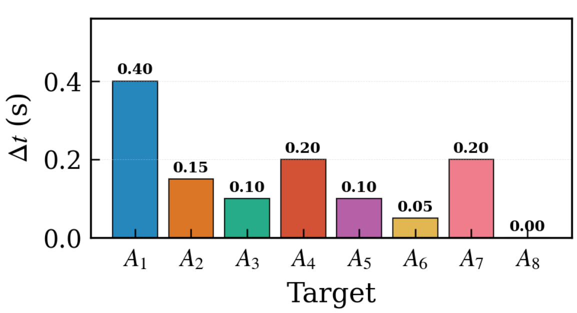


Fig. 25. Terminal arrival time differences Δt within each PN interceptor group in Case 1.

2) Case 2: Interception against Continuous Weaving Targets: Case 2 uses a sinusoidal penetration maneuver [?]. Its continuous horizontal and vertical LOS variation provides a more persistent disturbance than the Case 1 transition and tests whether group timing can be maintained while all three command channels respond.

1) ART-MAPPO: The three-dimensional paths in Fig. **??** show that all eight groups maintain spatial closure against the weaving targets. The horizontal projection in Fig. **??** exposes the lateral curvature without hiding the altitude evolution shown in the 3-D view.

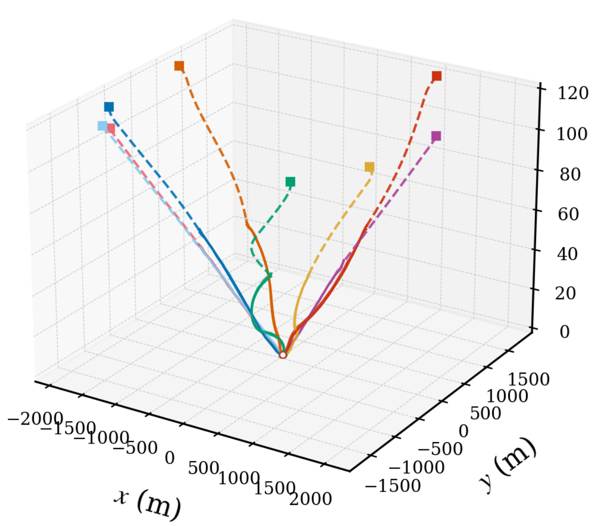


Fig. 26. Three-dimensional ART-MAPPO engagement trajectories against continuous weaving targets in Case 2.

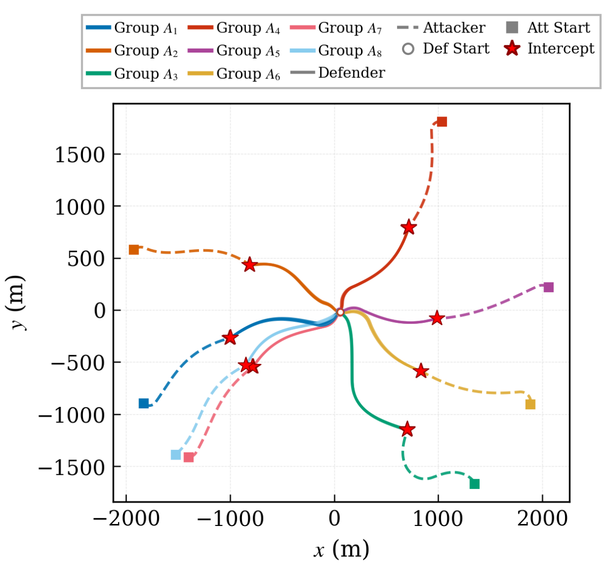


Fig. 27. Horizontal projection of the ART-MAPPO engagement trajectories in Case 2.

As shown in Figs. **??**–**??**, ART-MAPPO adapts n_y to the continuous lateral disturbance while using n_z to regulate vertical closure. The heading and pitch histories remain smooth relative to the command variations, and all load channels remain inside the prescribed envelope. The axial command raises or preserves closing speed without eliminating the control authority required in the yaw and pitch planes.

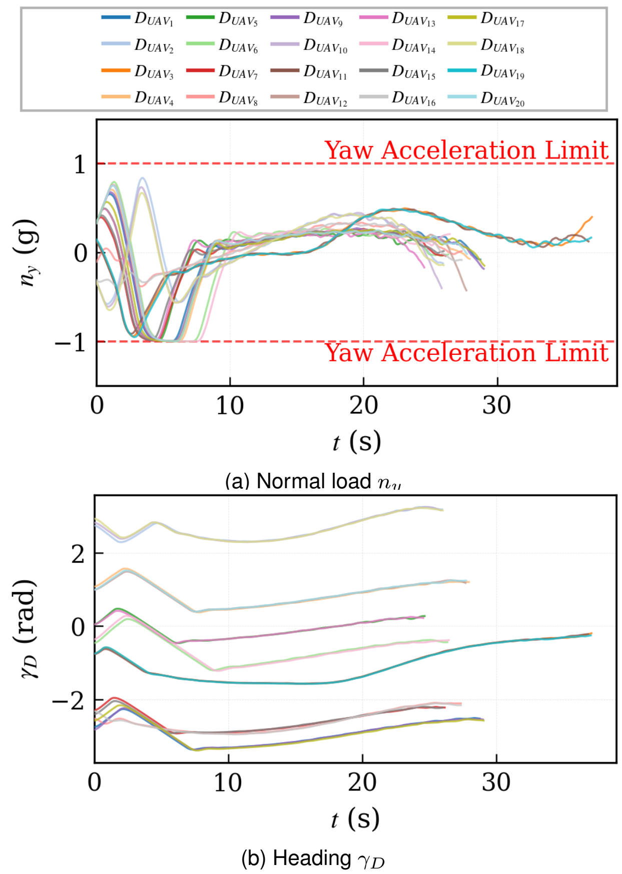


Fig. 28. Yaw-plane command and response of ART-MAPPO in Case 2.

Despite the continuous LOS disturbance, the ART-MAPPO t_{go} curves converge within their assigned groups (Fig. **??**). The largest terminal arrival-time difference is 0.30 s in Fig. **??**, demonstrating that the timing objective remains active in the three-dimensional engagement.

2) PN baseline: Figures **??** and **??** show that the constrained PN baseline also reaches all assigned targets in the selected Case 2 engagement. The spatial paths exhibit the reactive response to the continuously varying LOS, while the plan view and 3-D view together distinguish lateral curvature from vertical closure.

The PN profiles in Figs. **??**–**??** show sustained reactive changes in the horizontal channel together with the vertical and axial responses. Since all commands are clipped to the same envelope as ART-MAPPO, excursions that reach the plotted ± 1 lines represent saturation of available authority rather than an unrealizable command.

Figure **??** shows that the selected PN engagement also achieves groupwise time-to-go convergence. Its maximum arrival spread is 0.35 s (Fig. **??**), whereas ART-MAPPO gives 0.30 s. These representative runs therefore show successful three-dimensional interception for both methods and slightly tighter terminal timing for ART-MAPPO; the Monte Carlo

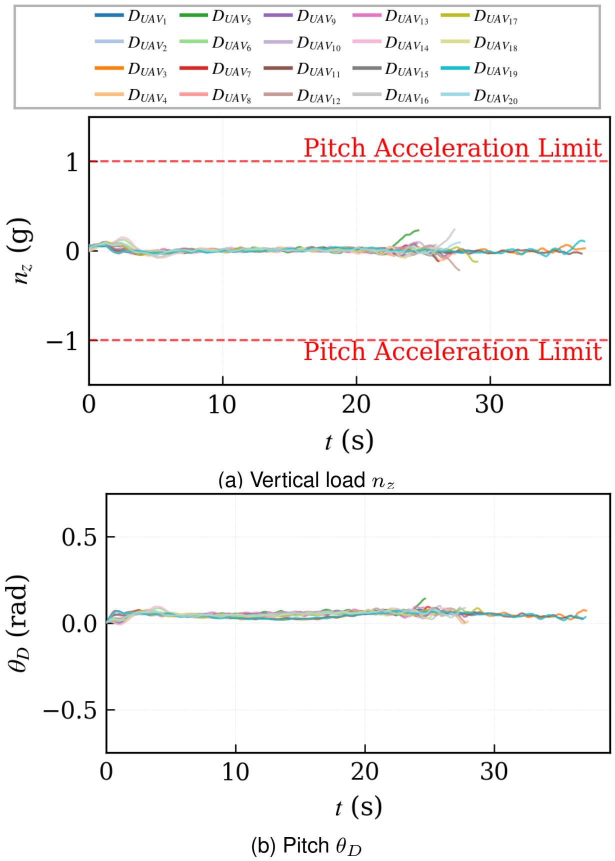


Fig. 29. Pitch-plane command and response of ART-MAPPO in Case 2.

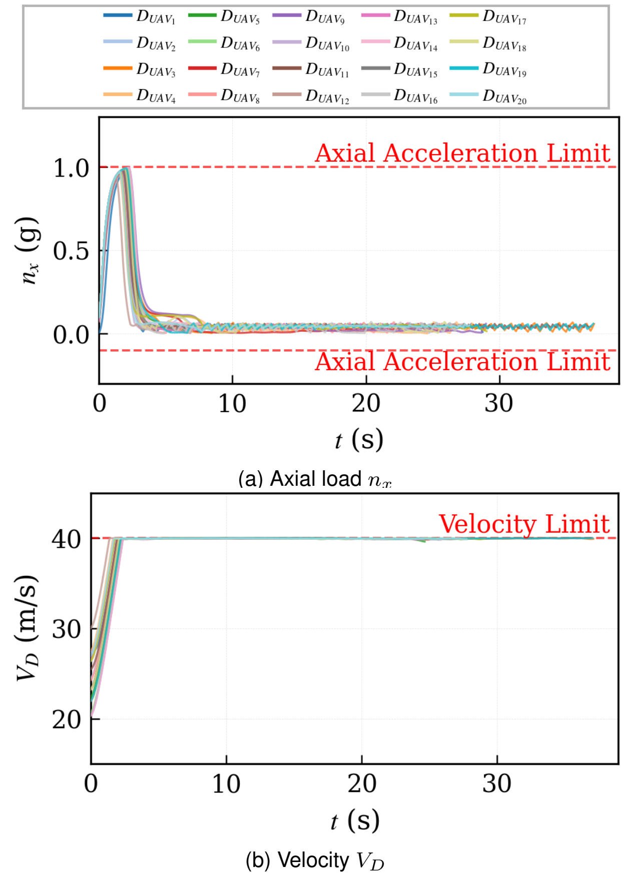


Fig. 30. Axial command and speed response of ART-MAPPO in Case 2.

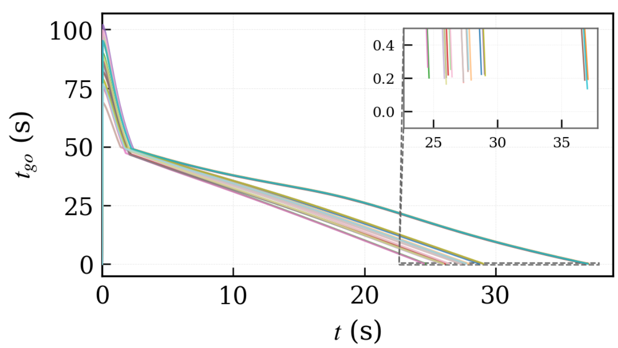


Fig. 31. Time-to-go (t_{go}) profiles of the defending swarm using ART-MAPPO in Case 2.

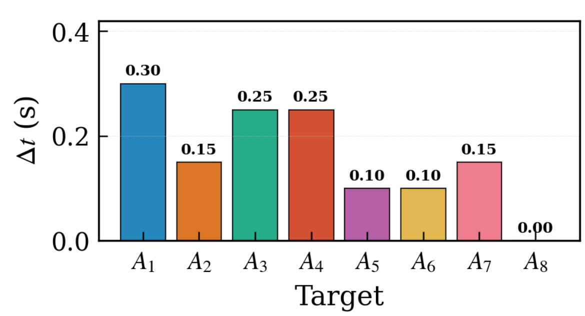


Fig. 32. Terminal arrival time differences Δt within each ART-MAPPO interceptor group in Case 2.

analysis below remains the basis for broader reliability comparisons.

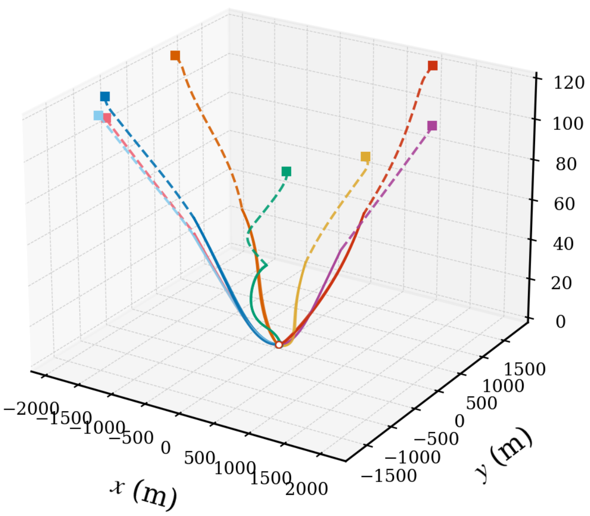


Fig. 33. Three-dimensional engagement trajectories of the defending swarm using the PN baseline in Case 2.

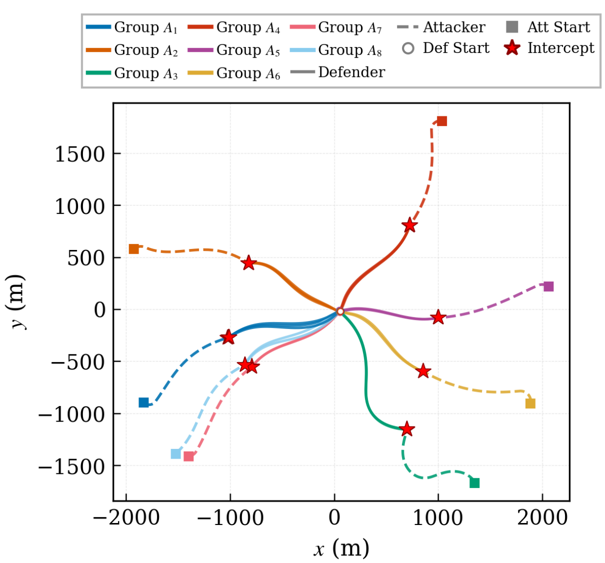
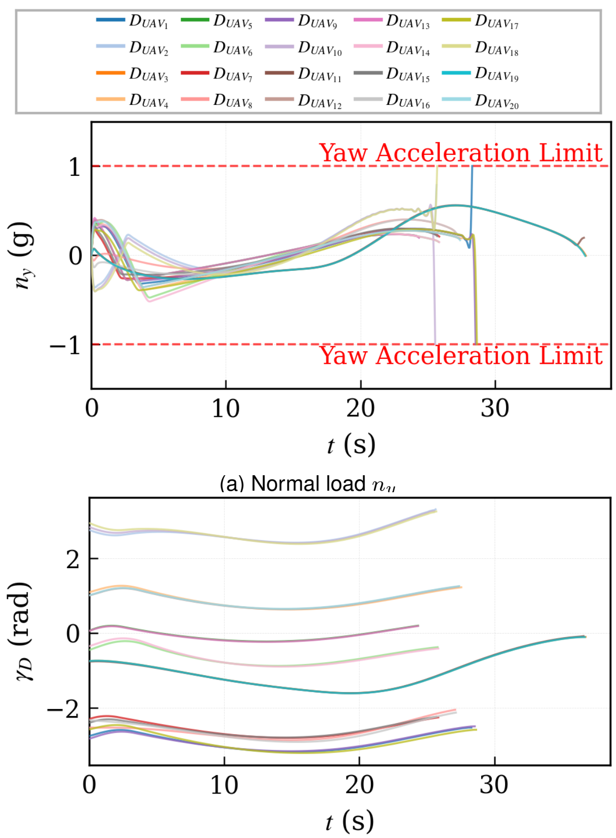


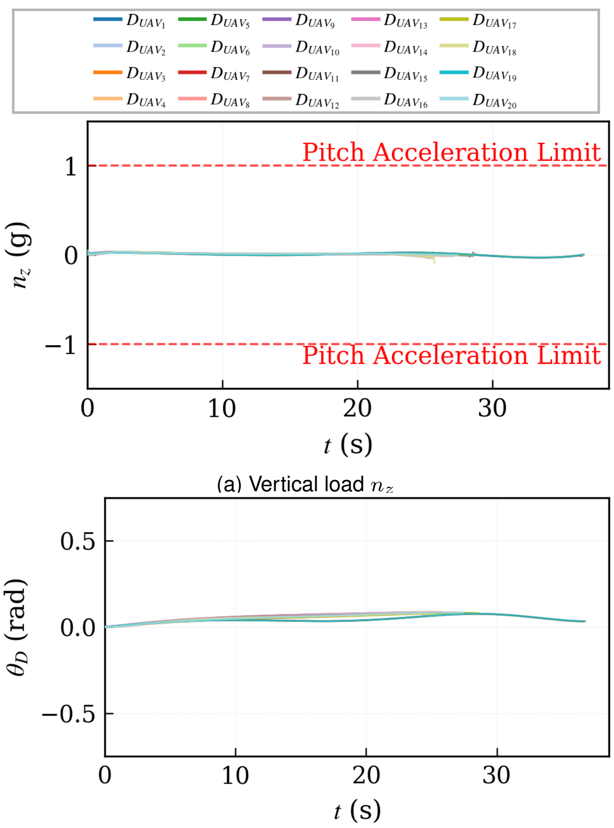
Fig. 34. Horizontal projection of the PN engagement trajectories in Case 2.



(a) Normal load n_y

(b) Heading γ_D

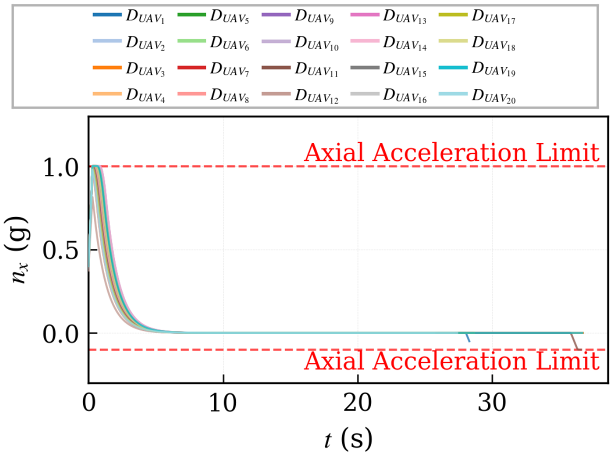
Fig. 35. Yaw-plane command and response of the capacity-matched PN baseline in Case 2.



(a) Vertical load n_z

(b) Pitch θ_D

Fig. 36. Pitch-plane command and response of the capacity-matched PN baseline in Case 2.



(a) Axial load n_x

(b) Velocity V_D

Fig. 37. Axial command and speed response of the capacity-matched PN baseline in Case 2.

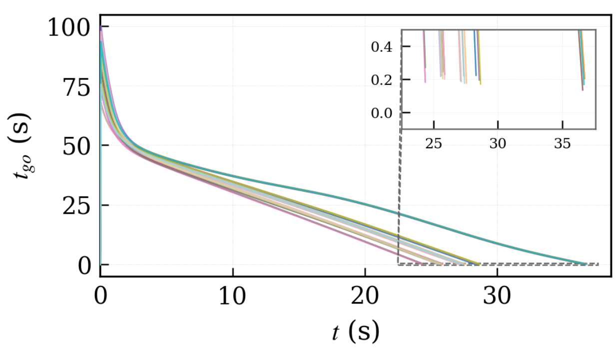


Fig. 38. Time-to-go (t_{go}) profiles of the defending swarm using the PN baseline in Case 2.

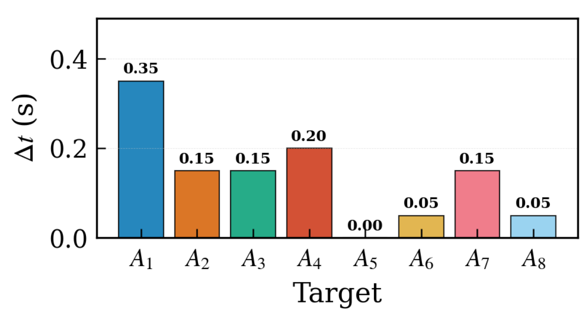


Fig. 39. Terminal arrival time differences Δt within each PN interceptor group in Case 2.

G. Monte Carlo Statistical Analysis and Robustness

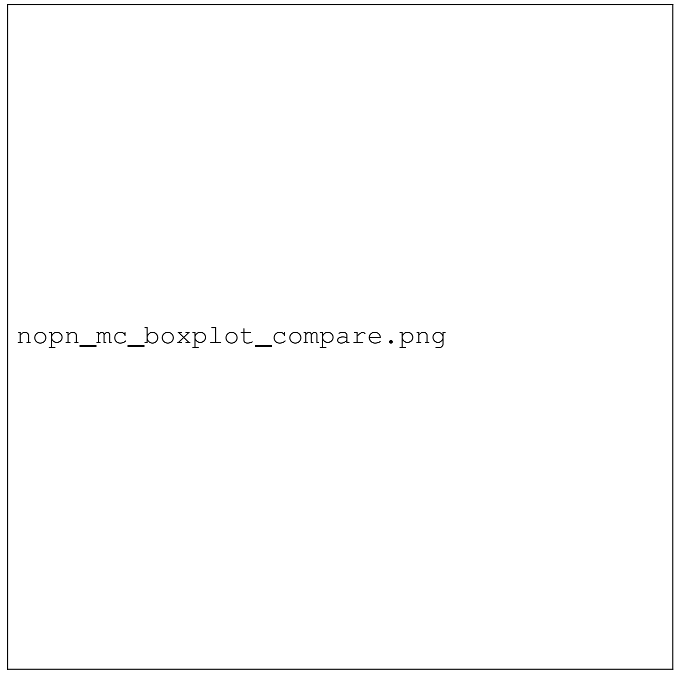
To rigorously quantify the robustness of the proposed framework, 1000 independent Monte Carlo trials are conducted for each case. It is important to note that the statistical distributions presented in Fig. **??** ($E_{co\text{-}time}$, E_n, E_{miss}, and E_t) are exclusively derived from the successful interception trials. Specifically, out of the 1000 trials in Case 1, ART-MAPPO and PN successfully intercepted the targets 987 and 944 times, respectively. In the highly dynamic Case 2, ART-MAPPO maintained a remarkable 971 successful interceptions, whereas the baseline PN succeeded only 175 times.

Based on the statistical distributions, three key findings emerge:

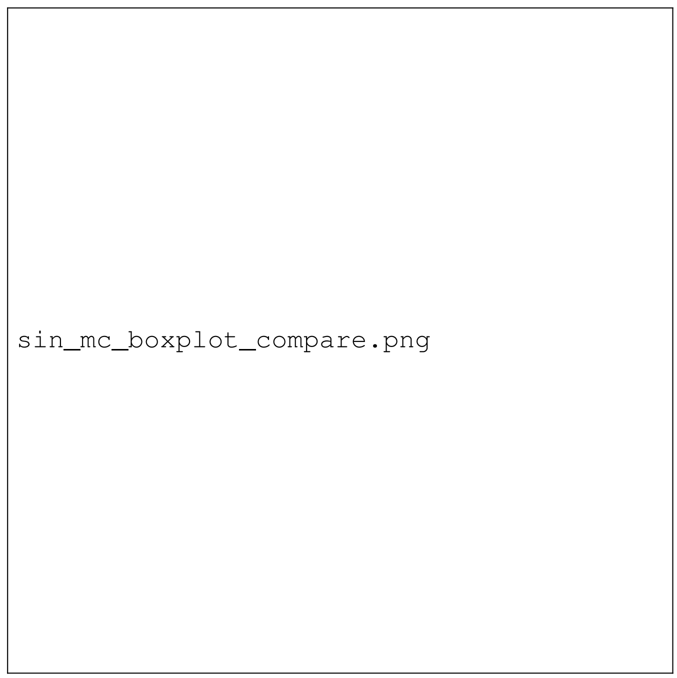
(i) Interception reliability and precision. The Interception Success Rates (ISRs) directly reflect the algorithms' adaptability to aggressive target maneuvers. While PN achieves a respectable 94.40% against the evasive target (Case 1), its performance precipitously degrades to 17.50% when facing the continuous weaving target (Case 2). Conversely, ART-MAPPO sustains exceptional ISRs of 98.70% and 97.14%, demonstrating profound robustness against continuous, high-frequency line-of-sight (LOS) oscillations. Furthermore, the E_{miss} boxplots confirm that ART-MAPPO consistently achieves a lower median miss distance with a significantly narrower interquartile range (IQR), ensuring lethal precision even under worst-case initializations.

(ii) Terminal required overload and operational efficiency. The metric E_n denotes the terminal required normal overload, which is a critical constraint for flight safety. As depicted in the boxplots, ART-MAPPO's E_n is tightly clustered around a safe median of $\approx 0.08\,\mathrm{g}$ (Case 1) and $\approx 0.20\,\mathrm{g}$ (Case 2). In stark contrast, particularly in Case 2, PN's E_n distribution is heavily squeezed near the $1.0\,\mathrm{g}$ upper bound. This exposes severe terminal actuator saturation, as PN struggles to reactively track the sinusoidal maneuvers. Additionally, ART-MAPPO strictly bounds the engagement duration (E_t) within a tight and predictable time window, reflecting superior forward energy management and rapid closure capabilities.

(iii) Robust temporal coordination. The metric $E_{co\text{-}time}$ evaluates the temporal synchronization of the defending swarm. Across the successful Monte Carlo trials in both highly dynamic scenarios, ART-MAPPO maintains a median $E_{co\text{-}time}$ well below $0.10\,\mathrm{s}$. This consistently tight distribution confirms that the proposed algorithm can reliably shape the swarm's velocity profile to satisfy the strict temporal requirements for simultaneous saturation strikes, effectively neutralizing the risk of sequential piecemeal defeat.



(a) Case 1



(b) Case 2

Fig. 40. Boxplots of Monte Carlo trials comparing ART-MAPPO and PN (data derived from successful interceptions).

VI. Conclusion

In this paper, the cooperative interception problem of UAV swarms against highly dynamic and maneuverable attacking UAV swarms has been investigated, explicitly accounting for real-time decision requirements, coordinated engagement effectiveness, and terminal overload risk. The problem is formulated by jointly considering target assignment and cooperative guidance, and an intelligent temporally coordinated interception framework is developed. A fast IDBO-based target assignment mechanism enables efficient and accurate resource allocation in dynamic environments. The ART-MAPPO cooperative guidance strategy under the CTDE paradigm improves learning stability and training efficiency in large-scale adversarial settings. Extensive three-dimensional simulations demonstrate accelerated target assignment, enhanced interception efficiency, improved interception success rates, and effective alleviation of terminal overload saturation.

For future work, promising directions include cooperative interception under incomplete information, dynamic target reassignment in unknown environments, time-constrained assignment with explicit completion deadlines, and integration with realistic constraints such as limited bandwidth, aerodynamic coupling, and heterogeneous UAV capabilities.

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TABLE VI
GROUPED NOTATION SUMMARY

Symbol	Description	Symbol	Description
Sets, indices, and physical quantities			
$\mathcal{D}, \mathcal{T}$	Defender and target sets.	N_D, N_A	Numbers of defenders and targets.
i, j, k, t	Defender, target, assignment/episode, and guidance-step indices.	$\mathcal{N}_i, \mathcal{G}_j$	Communication neighbors of defender i and defenders assigned to target j.
$\mathbf{p}_i, \mathbf{v}_i, V_i$	Position, velocity vector, and speed of defender i.	$\boldsymbol{\ell}_i, \gamma_{D_i}, \theta_{D_i}$	LOS unit vector, defender heading, and flight-path angle.
$d_i, t_{go,i}$	Defender–target range and estimated time-to-go.	$n_{x_i}, n_{y_i}, n_{z_i}$	Axial, horizontal-normal, and vertical-normal load-factor commands.
Target-assignment quantities			
$X_{ij}, \mathbf{X}$	Binary assignment variable and assignment matrix.	$L_{\max}$	Maximum defenders assigned to one target.
p_{ij}^{int}	Pairwise interception-probability proxy.	$\boldsymbol{\xi}_i, \pi_{ij}$	Latent and sigmoid-mapped target preferences.
u_{ij}, χ_{ij}	Target-specific local utility and combat advantage.	$b_{ij}, \mathcal{W}_{i,j}$	Target bid and defender i's local winner set for target j.
$\Gamma^{(k)}$	Pairwise assignment-consensus disagreement.	$N_{\text{pop}}, K_{\text{IDBO}}$	Local population size and number of IDBO iterations.
Learning quantities			
$\mathbf{o}_i^{(t)}, \mathbf{s}_t$	Local actor observation and joint critic state.	$\mathbf{H}_i, \mathbf{h}_i, \mathbf{g}_i$	Attention tokens, residual feature, and GRU state.
$\mathbf{a}_i^{(t)}, \mathcal{A}$	Three-dimensional load command and feasible action set.	π_θ, V_ϕ	Shared decentralized actor and centralized value function.
$\mathcal{T}_i, G_i^{(k)}$	Training-time trust state and episodic return.	$\varrho_t, \widehat{A}_t$	PPO importance ratio and GAE advantage estimate.
$\varepsilon_{\text{PPO}}, c_{\text{DC}}$	PPO clipping factor and dual-clip parameter.	$\delta_{\text{KL}}, \beta_{\text{KL}}$	Target KL divergence and adaptive KL penalty.
$r_i^{(t)}, \gamma_{\text{RL}}$	Per-step reward and RL discount factor.	H_{ep}	Episode horizon.
Evaluation quantities			
$t_i^{\text{arr}}, \mathcal{W}_i$	Arrival time and terminal measurement window.	$E_{\text{co-time}}, E_n$	Arrival-time dispersion and terminal-window load metric.
E_{miss}, E_t	Terminal miss distance and coordinated engagement duration.		

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APPENDIX A
NOTATION TABLE

Table **??** groups the main symbols by role. Symbols used only as temporary indices inside a single equation are explained locally and omitted here.